RESEARCH ARTICLE SUMMARY

NEUROIMMUNOLOGY

C1q and immunoglobulins mediate activity-dependent synapse loss in the adult brain

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INTRODUCTION: Synaptic elimination is a fundamental process that can engage microglia and the classical complement cascade, yet the mechanisms that specify which synapses are targeted by complement component 1q (C1q), the initiating component of this cascade, remain incompletely understood. Neuronal activity shapes synaptic connectivity, but whether it interfaces with complement pathways to regulate synapse loss in the adult brain, including in disorders such as Alzheimer's disease (AD), is unclear. In peripheral tissues, antigen-bound immunoglobulins can recruit C1q to targets, raising the possibility that adaptive immune mechanisms contribute to complement-dependent synaptic elimination in the central nervous system.

RATIONALE: We tested whether neuronal activity specifies complement-dependent synaptic elimination in the adult brain. Further, guided by unbiased spatial transcriptomic analyses, we investigated whether adaptive immune components contribute under conditions of altered circuit activity. To address this, we manipulated neuronal activity in defined hippocampal circuits in adult wild-type mice and examined the impact on synapses and neuroimmune interactions, with additional validation in an AD mouse model.

RESULTS: Inducing neuronal hyperactivity in adult wild-type mice, using chemogenetic and pharmacological approaches, led to region- and input-specific loss of synaptic terminals in a C1q-dependent manner. Conversely, reducing neuronal hyperactivity in an AD mouse model decreased C1q deposition and partially restored synaptic density, supporting a link between activity state and complement-associated synapse loss in both physiological and disease-relevant contexts.

Spatial transcriptomic analyses in the chemogenetically activated wild-type brains identified activity-dependent accumulation of B lymphocyte lineage cells, including antibody-secreting cells, in leptomeningeal niches adjacent to regions of elevated activity. B cell receptor sequencing and photoconversion experiments supported their origin in the dura and localization to activity-modulated hippocampal regions. At the synaptic level, super-resolution imaging demonstrated association of immunoglobulin M (IgM) with C1q-localized synaptic terminals. Genetic perturbation approaches further indicated that secreted IgM and antigen specificity contribute to activity-dependent C1q deposition and synapse loss.

CONCLUSION: These findings identify neuronal activity as a key regulator of complement-dependent synaptic elimination in the adult brain. In addition, they support a contribution of adaptive immune components in the hyperactivity-induced wild-type brain, whereby B lymphocyte lineage cells and IgM participate in C1q-associated synaptic loss. Together, this work supports a model in which activity state interfaces with innate immune mechanisms to mediate synaptic elimination, with adaptive immune components providing an additional modulatory layer. □

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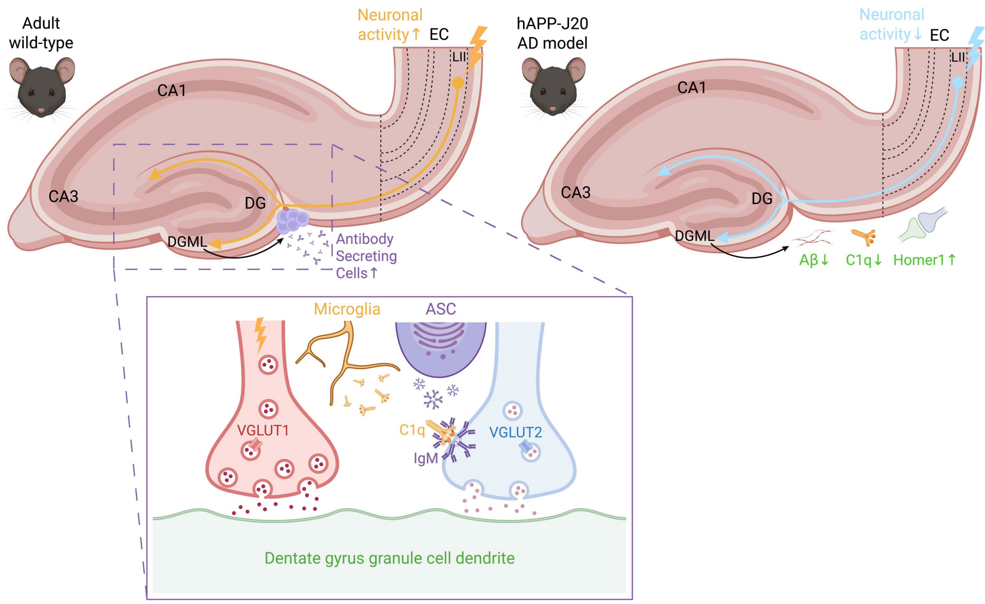

Neuronal activity modulation induces synaptic refinement through innate and adaptive immune components. Enhancement of perforant pathway neuronal activity leads to loss of nonactivated vesicular glutamate transporter 2 (VGLUT2) presynaptic terminals. In this context, synapse loss is coordinated by release of both C1q from microglia and antigen-specific IgM from antibody-secreting B-lineage cells recruited to the hippocampus. In contrast, inhibition of perforant pathway neuronal activity induces a reduction of amyloid-β (Aβ) and C1q to restore Homer1 postsynaptic terminals. Created with BioRender.com. CA, cornu ammonis; DG, dentate gyrus; DGML, dentate gyrus molecular layer; EC, entorhinal cortex; ASC, antibody-secreting cell.

NEUROIMMUNOLOGY

C1q and immunoglobulins mediate activity-dependent synapse loss in the adult brain

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Complement component 1q (C1q), the initiator of the classical complement cascade, mediates synaptic elimination in development and disease, yet the triggers for its deposition on synapses remain unclear. Using in vivo chemogenetics, we demonstrate that neuronal hyperactivity induces region-specific, C1q-dependent synapse loss in the adult hippocampus. Suppressing perforant pathway hyperactivity in a mouse model of Alzheimer's disease reduced local amyloid-β amounts and C1q deposition and partially rescued synapse loss. Combining spatial transcriptomics, live cell tracking, and super-resolution microscopy, we identified association of antibody-secreting B-lineage cells in the adult hippocampus with activity-dependent, C1q-mediated synapse loss under physiological conditions. Together, these findings link neuronal hyperactivity to C1q-mediated synapse loss in the adult brain and implicate immunoglobulins as players in this process.

Disruptions in neural circuit activity are a hallmark of early Alzheimer's disease (AD) (*1*), yet the mechanisms linking pathological activity to selective synapse loss remain poorly understood. Genetic studies implicate immune pathways in AD (*2*) and across multiple neurodegenerative models, the innate immune protein C1q (complement component 1q) accumulates at synapses in vulnerable brain regions and is required for their elimination (*3*–*9*). However, what initiates C1q deposition in the adult brain remains a fundamental unanswered question.

Region-specific neuronal hyperactivity emerges early in both individuals with AD and mouse models (*10*, *11*). However, whether hyperactivity itself is sufficient to engage complement and mediate synaptic remodeling in the healthy brain is unknown. We show that neuronal hyperactivity is a spatially precise trigger of complement-mediated synapse loss in the adult hippocampus. Using chemogenetic modulation in wild-type (WT) mice and the human amyloid precursor protein (hAPP)–J20 AD mouse model (J20) (*12*), we found that increased perforant pathway activity induces C1q deposition and synapse loss, whereas dampening hyperactivity reduces local amyloid-β (Aβ), diminishes C1q accumulation, and partially restores synaptic density. These results establish a link between circuit hyperactivity and complement-mediated synaptic elimination.

We identified a previously unrecognized immunological component of this process: B lymphocyte lineage antibody-secreting cells (ASCs) that localize to hippocampal regions receiving activated synaptic projections. Super-resolution imaging showed immunoglobulin M (IgM) juxtaposed to C1q-positive vesicular glutamate transporter 2 (VGLUT2) presynaptic terminals, suggesting that locally produced immunoglobulins might facilitate C1q deposition and synapse loss in WT mice. Collectively, our findings indicate a functional interaction between neuronal activity, immunoglobulins, and C1q-mediated synapse loss in the adult brain.

Results

Neuronal activity mediates C1q deposition and synapse loss

Neuronal activity regulates synaptic pruning during development (*13*). We first asked whether increased neuronal activity is sufficient to trigger synaptic C1q deposition in adult (3-month-old) WT mice. We targeted the perforant pathway connecting medial entorhinal cortex (MEC) layer II to the hippocampal dentate gyrus molecular layer (DGML; figs. S1 and S2). Mice received unilateral stereotaxic injection into the MEC of AAV8-CaMKIIa-HA-hM3Dq-IRES-mCitrine, an excitatory DREADD that is activated by clozapine *N*-oxide (CNO) (*14*). The DGML projection site lies >3.5 mm away from the injection site, minimizing surgical confounds; unilateral targeting further enabled within-animal comparison with the contralateral hemisphere (fig. S1C) (*15*). Independent control groups included AAV8-CaMKIIa-EGFP–injected mice, and all cohorts received daily saline or CNO intraperitoneal injections for five consecutive days, with brains harvested 90 min after the final CNO injection (Fig. 1A).

CNO treatment led to increased c-Fos expression in the ipsilateral MEC compared with the contralateral side in hM3Dq-expressing mice (hM3Dq+CNO), suggesting elevated neuronal activity (Fig. 1, B and C). No differences in c-Fos were observed in any of the three control groups: hM3Dq virus with saline, EGFP (enhanced green fluorescent protein) control virus with CNO, or EGFP control virus with saline (Fig. 1C). Head-fixed recordings with Neuropixels 1.0 probes confirmed an increase in the activity of units in the transduced (GFP-positive) areas of the MEC (fig. S3). At the synaptic projection site of hM3Dq+CNO mice, the ipsilateral hippocampal DGML exhibited increased C1q deposition compared with the contralateral DGML, whereas we did not find interhemispheric differences in control groups (Fig. 1, D and E). Increased C1q deposition was anatomically restricted to the activated projection zone and absent in nontargeted regions including the DG hilus, striatum, and cerebellum (figs. S4 and S5). These data suggest that increased neuronal activity selectively enhances C1q deposition at activated synaptic termination sites.

We next sought to determine whether activity modulation alters synaptic density at the projection site using Airyscan super-resolution

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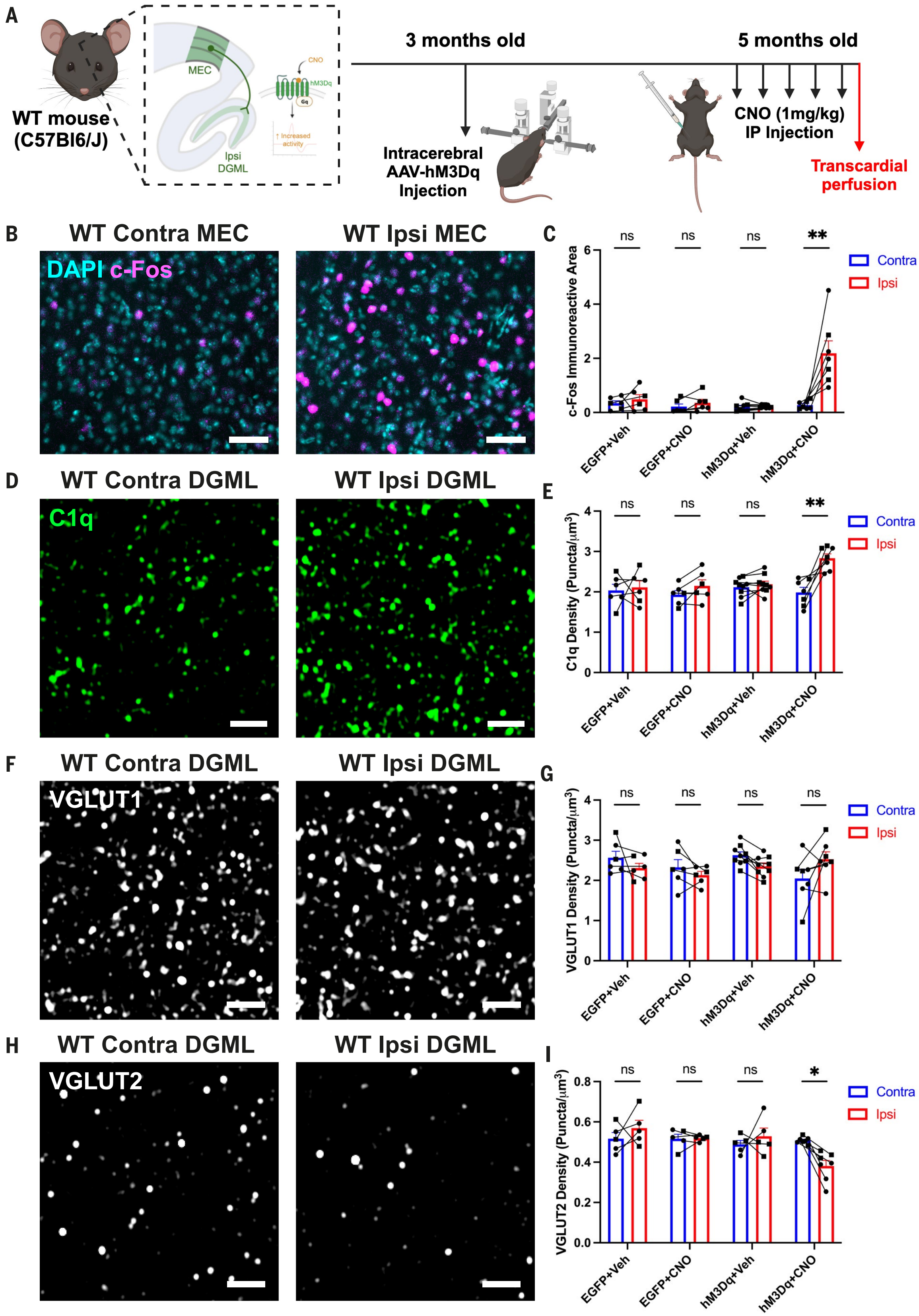

Fig. 1. Increased neuronal activity alters C1q deposition and presynaptic density in WT mice. (**A**) Schematic of WT MECLII-DGML experimental paradigm. Created with BioRender.com. IP, intraperitoneal. (**B**) Representative widefield images of c-Fos labeling in contralateral (Contra) and ipsilateral (Ipsi) MEC of hM3Dq+CNO WT mouse. Scale bars, 50 μm. (**C**) Quantification of c-Fos–immunoreactive area between hemispheres in each mouse group. $n = 6$ EGFP+Veh mice, $n = 5$ EGFP+CNO mice, $n = 8$ hM3Dq+Veh mice, and $n = 7$ hM3Dq+CNO mice, two or three brain sections per hemisphere, three ROIs analyzed per section. P values from Bonferroni post hoc test after two-way mixed analysis of variance (ANOVA) with group as between-subject and hemisphere as within-subject factor (interaction $P = 0.008$ after $\log_{10}$ transformation). (**D**) Representative

Airyscan images of C1q puncta at contralateral and ipsilateral DGML of hM3Dq+CNO WT mouse. Scale bars, 2 μm. (**E**) Quantification of C1q puncta density between hemispheres in each mouse group. $n = 6$ EGFP+Veh mice, $n = 6$ EGFP+CNO mice, $n = 9$ hM3Dq+Veh mice, and $n = 7$ hM3Dq+CNO mice, one to three brain sections per hemisphere, three ROIs analyzed per section. *P* values from Bonferroni post hoc test after two-way mixed ANOVA with group as between-subject and hemisphere as within-subject factor (interaction $P = 0.004$). (**F**) Representative Airyscan images of VGLUT1 puncta at contralateral and ipsilateral DGML of hM3Dq+CNO WT mouse. Scale bars, 2 μm. (**G**) Quantification of VGLUT1 puncta density between hemispheres in each mouse group. $n = 6$ EGFP+Veh mice, $n = 6$ EGFP+CNO mice, $n = 9$ hM3Dq+Veh mice, and $n = 7$ hM3Dq+CNO mice, one to three brain sections per hemisphere, three ROIs analyzed per section. *P* values from Bonferroni post hoc test after two-way mixed ANOVA with group as between-subject and hemisphere as within-subject factor (interaction $P = 0.019$). (**H**) Representative Airyscan images of VGLUT2 puncta at contralateral and ipsilateral DGML of hM3Dq+CNO WT mouse. Scale bars, 2 μm. (**I**) Quantification of VGLUT2 puncta density between hemispheres in each mouse group. Data are mean ± SEM, $n = 5$ EGFP+Veh mice, $n = 5$ EGFP+CNO mice, $n = 5$ hM3Dq+Veh mice, and $n = 7$ hM3Dq+CNO mice, two to four brain sections per hemisphere, two or three ROIs analyzed per section. *P* values from Bonferroni post hoc test after two-way mixed ANOVA with group as between-subject and hemisphere as within-subject factor (interaction $P = 0.015$). Throughout, square points represent males and circular points represent females. Linked points indicate data from the same mouse brain. One point is equal to one hemisphere average across brain sections. Data are mean ± SEM. $^{ns}P > 0.05$, $*P < 0.05$, $**P < 0.01$, $***P < 0.001$.

confocal microscopy. Specifically, we quantified VGLUT1- and VGLUT2-immunoreactive presynaptic puncta in the DGML, where perforant pathway terminals are predominantly VGLUT1-positive (fig. S4) (*16*). VGLUT1-immunoreactive presynaptic puncta density did not differ between hemispheres in any group analyzed (Fig. 1, F and G). In contrast, hM3Dq+CNO treatment selectively reduced VGLUT2 terminal density at the ipsilateral DGML compared with the contralateral side (Fig. 1, H and I). Postsynaptic Homer1 amounts remained unchanged between hemispheres (fig. S6, A and B). Akin to C1q deposition, the loss of VGLUT2-immunoreactive synaptic puncta was confined to the ipsilateral DGML (activated projection zone) and absent from control regions including the DG hilus, striatum, and cerebellum (fig. S6, C to F), consistent with local refinement of neighboring VGLUT2 presynaptic terminals.

To determine whether the observed activity-dependent C1q and VGLUT2 loss generalizes to other pathways, we activated dorsal cornu ammonis 1 (CA1) projections to MEC layer V (fig. S7, A to C). CNO treatment induced strong c-Fos expression in hM3Dq-expressing CA1 pyramidal neurons (fig. S7, D and E) and increased C1q deposition (fig. S7, F and G) alongside reduced VGLUT2 density (fig. S7, H and I) at the targeted layer V site, mirroring the effects observed in the perforant pathway (Fig. 1). These data suggest that activity-dependent C1q deposition and selective VGLUT2 synapse loss represent a generalizable mechanism across corticohippocampal circuits.

Dampening neuronal hyperactivity reduces C1q deposition and ameliorates synapse loss in an AD mouse model

We next performed a complementary experiment to confirm the functional link between neuronal activity, C1q expression, and synapse health. In J20 mice, neuronal hyperactivity in the cortex is detected from 4 months old, before amyloid plaque pathology (*11*, *17*–*19*). We previously showed at this age that there is elevated hippocampal C1q deposition with Homer1-immunoreactive postsynaptic loss (fig. S8, A to D) (*3*). We targeted the perforant pathway with an inhibitory DREADD (AAV8-CaMKIIa-hM4Di-HA-IRES-mCitrine) and administered CNO daily for 5 days (Fig. 2A), including three control groups to account for viral injection and CNO administration, paralleling the WT paradigm.

In hM4Di+CNO J20 mice, phosphorylated pyruvate dehydrogenase (pPDH), a marker of reduced neuronal activity (*20*), increased at the ipsilateral MEC relative to the contralateral side (Fig. 2, B and C); there were no differences in pPDH amounts between hemispheres in controls (Fig. 2C). Head-fixed recordings with Neuropixels 1.0 probes demonstrated a decrease in the activity of units in the transduced (GFP-positive) areas of the MEC after CNO-mediated DREADD activation (fig. S9).

Consistent with decreased neuronal activity, C1q deposition was reduced in the ipsilateral DGML of hM4Di+CNO J20 group (Fig. 2, D and E), and Homer1-immunoreactive postsynaptic puncta were increased (Fig. 2, F and G). These changes were absent in control J20 mouse groups (Fig. 2, E and G), indicating that these changes in C1q and Homer1 puncta are associated with DREADD+CNO-dependent modulation of neuronal activity. These data in the J20 mice together indicate that dampening neuronal hyperactivity ameliorates C1q deposition and synapse loss in J20 mice, providing a reciprocal proof of principle for their functional relationship, as demonstrated in the WT mice.

Aβ production scales with synaptic activity (*21*–*24*); therefore we examined whether reducing neuronal hyperactivity in J20 mice altered local Aβ amounts. Indeed, both 4G8 and 6E10 staining (to label endogenous murine and transgenic human Aβ) showed decreased punctate Aβ immunoreactivity at the ipsilateral DGML of hM4Di+CNO J20 mice (Fig. 2, H and I), indicating that activity suppression lowers local Aβ production at the synapse. However, Bassoon+Homer1 colocalized synaptic puncta showed no difference ($P = 0.06$; fig. S8, G and H), suggesting no mitigation of synapse loss. Here, we used Bassoon instead of VGLUT1 or VGLUT2 to provide a broader assessment of presynaptic integrity, particularly at the active zone (*25*). These data demonstrate that dampening neuronal hyperactivity might partially protect synapses; however, additional experiments are needed to support this hypothesis. Together with the WT experiments, these data support a functional link between neuronal activity, C1q deposition, and synapse loss.

C1q is required for activity-dependent synapse loss in WT mice

A higher percentage of VGLUT2 puncta colocalized with C1q in the ipsilateral DGML of hM3Dq+CNO mice compared with the contralateral side (Fig. 3, A and B, and fig. S10A), whereas the percentage of VGLUT1 terminals associated with C1q did not change (fig. S10, B and C), suggesting selective C1q "tagging" of VGLUT2 synapses. To test whether C1q is required for activity-dependent synapse loss, we chemogenetically activated the perforant pathway in *C1qa* knockout (KO) mice (Fig. 3C) (*26*). As in WT mice (Fig. 1, B and C), hM3Dq-expressing *C1qa* KO mice showed increased c-Fos expression at the MEC after CNO treatment (Fig. 3, D and E), suggesting robust DREADD-dependent activity of the MEC neurons. However, unlike WT mice (Fig. 1, H and I), *C1qa* KO mice exhibited no activity-dependent difference in VGLUT2-immunoreactive puncta in the DGML between hemispheres (Fig. 3, F and G). These data suggest that C1q is required for activity-dependent synapse loss in adult WT brains.

In the adult mouse brain, C1q is primarily produced by microglia (*27*). Single-molecule fluorescence in situ hybridization (smFISH/RNAScope) combined with immunohistochemistry (IHC) confirmed *C1qa* transcripts in P2Y12-immunoreactive microglia (fig. S11A). However, *C1qa* mRNA amounts were unchanged between hemispheres of chemogenetically activated WT mice (fig. S11, A and B). Although there could be more subtle or cell state–specific changes, these data show that there is no measurable activity-dependent increase of *C1qa* transcription, suggesting that activity induces C1q up-regulation at the protein level rather than transcriptionally. Microglial markers P2Y12, CD68 (Fig. 3,

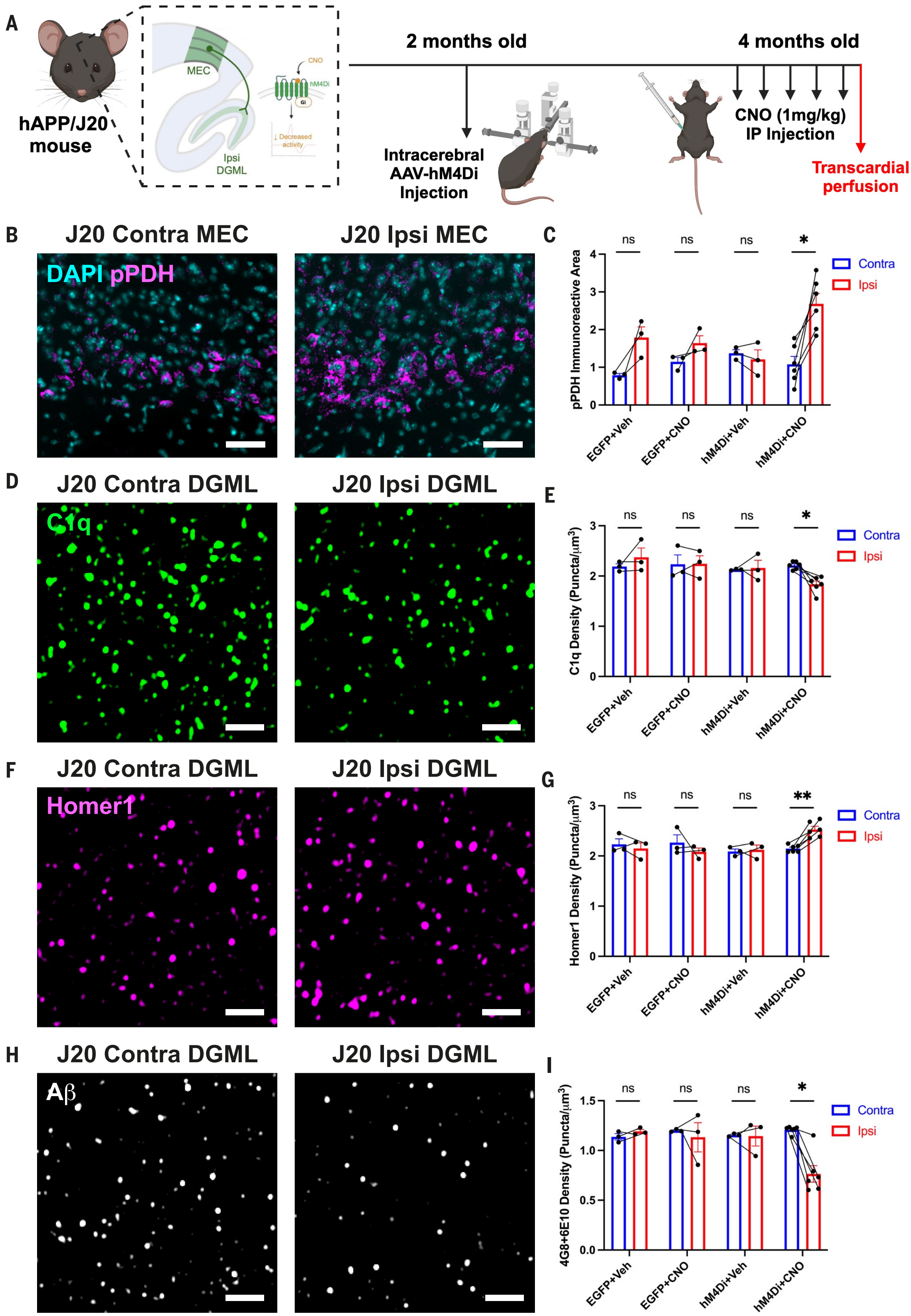

Fig. 2. Reduced neuronal activity ameliorates C1q deposition and postsynaptic loss in an AD mouse model. (**A**) Schematic of J20 MECLII-DGML experimental paradigm. Created with BioRender.com. (**B**) Representative widefield images of pPDH labeling at contralateral and ipsilateral MEC of hM4Di+CNO J20 mouse. Scale bars, 50 μm. (**C**) Quantification of pPDH immunoreactive area between hemispheres in each mouse group. $n = 3$ EGFP+Veh mice, $n = 3$ EGFP+CNO mice, $n = 3$ hM4Di+Veh mice, and $n = 6$ hM4Di+CNO mice, two or three brain sections per hemisphere, three ROIs analyzed per section. P values from Bonferroni post hoc test after two-way mixed ANOVA with group

as between-subject and hemisphere as within-subject factor (interaction $P = 0.020$). (**D**) Representative Airyscan images of C1q puncta at contralateral and ipsilateral DGML of hM4Di+CNO J20 mouse. Scale bars, 2 μm. (**E**) Quantification of C1q puncta density between hemispheres in hM4Di+CNO J20 mouse. $n = 3$ EGFP+Veh mice, $n = 3$ EGFP+CNO mice, $n = 3$ hM4Di+Veh mice, and $n = 6$ hM4Di+CNO mice, two brain sections per hemisphere, three ROIs analyzed per section. P values from Bonferroni post hoc test after two-way mixed ANOVA with group as between-subject and hemisphere as within-subject factor (interaction $P = 0.025$). (**F**) Representative Airyscan images of Homer1 puncta at contralateral and ipsilateral DGML of hM4Di+CNO J20 mouse. Scale bars, 2 μm. (**G**) Quantification of Homer1 puncta density between hemispheres in hM4Di+CNO J20 mouse. $n = 3$ EGFP+Veh mice, $n = 3$ EGFP+CNO mice, $n = 3$ hM4Di+Veh mice, and $n = 6$ hM4Di+CNO mice, two brain sections per hemisphere, two or three ROIs analyzed per section. P values from Bonferroni post hoc test after two-way mixed ANOVA with group as between-subject and hemisphere as within-subject factor (interaction $P = 0.014$). (**H**) Representative Airyscan images of 4G8+6E10 puncta (Aβ) at contralateral and ipsilateral DGML of hM4Di+CNO J20 mouse. Scale bars, 2 μm. (**I**) Quantification of 4G8+6E10 puncta (Aβ) density between hemispheres in hM4Di+CNO J20 mouse. $n = 3$ EGFP+Veh mice, $n = 3$ EGFP+CNO mice, $n = 3$ hM4Di+Veh mice, and $n = 6$ hM4Di+CNO mice, two brain sections per hemisphere, three ROIs analyzed per section. P values from Bonferroni post hoc test after two-way mixed ANOVA with group as between-subject and hemisphere as within-subject factor (interaction $P = 0.008$). Throughout, square points represent males and circular points represent females. Linked points indicate data from the same mouse brain. One point is equal to one hemisphere average across brain sections. Data are mean ± SEM. $^{ns}P > 0.05$, $*P < 0.05$, $**P < 0.01$, $***P < 0.001$.

H to J, and fig. S11, C to E), Iba1, and CD11b (fig. S11, F to H) did not differ between hemispheres at the DGML, indicating that microglia are not overtly activated. Thus, other activity-dependent changes likely facilitate increased C1q localization at hippocampal synapses at the activated synaptic projection site.

Immunoglobulin-producing B lymphocyte lineage cells are found at the site of activity-dependent synapse loss

To identify molecular changes underpinning activity-dependent C1q upregulation, we performed Visium spatial transcriptomics on hM3Dq+CNO mice. Consistent with our immunostaining findings (Fig. 3, H to J, and fig. S11), overall transcriptomic differences between the ipsilateral and contralateral DGML were minimal (Fig. 3K). The most up-regulated transcripts in the ipsilateral DGML were *Igha* and *Igkc* (Fig. 3, K and L, and fig. S12, A and B; most down-regulated were *Enpp2* and *Ttr*, fig. S13, A and B), alongside *Ighm* and *Jchain* (fig. S13, C and D), suggesting the presence of immunoglobulin-secreting cells.

smFISH and IHC confirmed the presence of *Igha*+B220+ immunoreactive cells (Fig. 3M and fig. S13E) and *Igkc*-expressing cells (fig. S13F) in the activated ipsilateral hippocampus, with few such cells detected contralaterally. Flow cytometry further demonstrated a higher number of CD19+B220+ B cells in the hippocampal tissue of hM3Dq+CNO mice compared with hM3Dq+vehicle controls (Fig. 3N and fig. S14A). We did not observe any differences in CD19+B220+ B cells in nonhippocampal brain tissue or dural meninges (fig. S14, B and C). Moreover, brief intravascular anti-CD45 labeling, a widely used method to distinguish circulating from noncirculating immune cells (*28*–*31*), did not stain the hippocampal B cells (fig. S14D), suggesting a noncirculating identity. These data indicate that B-lineage cells accumulate at hippocampal sites of enhanced neuronal activity.

Activity-dependent recruitment of ASCs to hippocampal niches

Given the limited resolution of Visium combined with potential lateral diffusion of transcripts (*32*), we applied Xenium spatial transcriptomics to provide single cell–level characterization of sections from three independent WT hM3Dq+CNO mice. We found *Igkc*-expressing cells predominantly in the ipsilateral hippocampus (Fig. 4, A and B), coexpressing multiple immunoglobulin heavy-chain transcripts, including *Igha*, *Ighg2b*, and *Ighm*, as well as *Jchain* (Fig. 4B). These cells also expressed transcription factors (*Prdm1*, *Xbp1*, *Irf4*) and cell surface markers (*Sdc1*, *Cd79a*, *Cd79b*, *Slamf7*), collectively indicating a plasmablast or plasma cell identity (Fig. 4B). Using hM3Dq-injected Aicda-CreERT2/R26ReYFP mice, which enable lineage tracing of B cells activated during tamoxifen exposure (*33*, *34*), we detected EYFP+CD138+ cells at the activated hippocampus following CNO treatment (fig. S15). This finding, suggests that a proportion of hippocampal B-lineage cells are likely recently antigen-stimulated ASCs.

Immunostaining confirmed IgA+CD138+, IgG+, and IgM+ cells at the interface of the hippocampal DG, subiculum, and the ventricular space; near the hippocampal fissure; and within the leptomeningeal velum interpositum (VI) adjacent to the activated DG (Fig. 4C and fig. S13G) (*35*). Furthermore, we found CD138+ cells nearby activated GFP-positive terminals at MEC layer V in the alternative CA1-MEC chemogenetic paradigm (fig. S7C), and Airyscan imaging showed that some IgG+ cells were positioned near vasculature, juxtaposed to GLUT1+ endothelial cells in the hippocampus (fig. S13H). Ig+ cells diminished in quantity at the ipsilateral hippocampus over a 2-week period post-CNO treatment (fig. S13I), whereas IgA and IgG immunofluorescence did not differ between hemispheres in control groups immediately after CNO administration (fig. S13, J and K).

To define their microanatomical localization, we labeled PECAM1+ endothelial cells, CRABP2+ leptomeningeal cells, and AQP4+ astrocyte endfeet, performing high-resolution confocal imaging and three-dimensional (3D) reconstruction (Fig. 4D). Most of the IgG+ cells were located at brain interfaces including within the leptomeningeal VI, with a smaller proportion situated in parenchymal regions (Fig. 4, E and F). We observed very few cells in the ventricular spaces, and despite their perivascular proximity, we did not find any within the PECAM1+ vascular lumen or perivascular compartment, defined as localization between PECAM1+ endothelial cells and AQP4+ astrocyte endfeet in 3D surface-rendered image reconstructions (Fig. 4, E and F).

Neither our spatial transcriptomic data nor IHC data suggest any gross blood-brain barrier (BBB) disruption in hM3Dq+CNO mice (Fig. 3, K and L, and fig. S9); we confirmed this by administering intravenous dextrans of varying molecular weights (10, 70, and 2000 kDa). No differences in dextran fluorescence were detected between ipsilateral and contralateral hippocampi in either hM3Dq+Veh or hM3Dq+CNO mice (fig. S16), indicating that the ipsilateral accumulation of Ig+ cells in hM3Dq+CNO mice is not attributable to DREADD injection or nonspecific BBB leakage. These results indicate that ASCs—plasmablasts and plasma cells—of the B lymphocyte lineage localize to leptomeningeal and parenchymal areas of the hippocampus in a regulated, activity-dependent manner.

To investigate the origin of the B lymphocyte lineage cells and the immunoglobulins they produce, we performed B cell receptor sequencing (BCR-seq) of dural meningeal, hippocampal, and striatal (control region) tissues from hM3Dq+CNO mice (Fig. 4G). A very high proportion of BCRs from both hippocampus and striatum were shared with dura, suggesting that brain B cells likely originate from the dura (Fig. 4H). The hippocampal BCR repertoire included IgA, IgG, and IgM heavy chains (Fig. 4I), in agreement with our Xenium and IHC findings (Fig. 4, B and C). To directly test whether hippocampal B lymphocyte lineage cells originate from the dural meninges or skull bone marrow (*36*, *37*), we used Kaede transgenic mice, which allow the tracking of photoconverted cells after ultraviolet (UV) light exposure (Fig. 4J) (*38*). Upon UV illumination of the skull, we detected a higher proportion of photoconverted CD19+B220^{lo}SCA1+ putative plasmablasts in the hippocampus of the hM3Dq+CNO Kaede mice compared with hM3Dq+Veh Kaede controls (Fig. 4K and fig. S14E). In contrast, the proportions of photoconverted plasmablasts in the dural meninges or in nonhippocampal brain regions did not differ between groups (fig. S14,

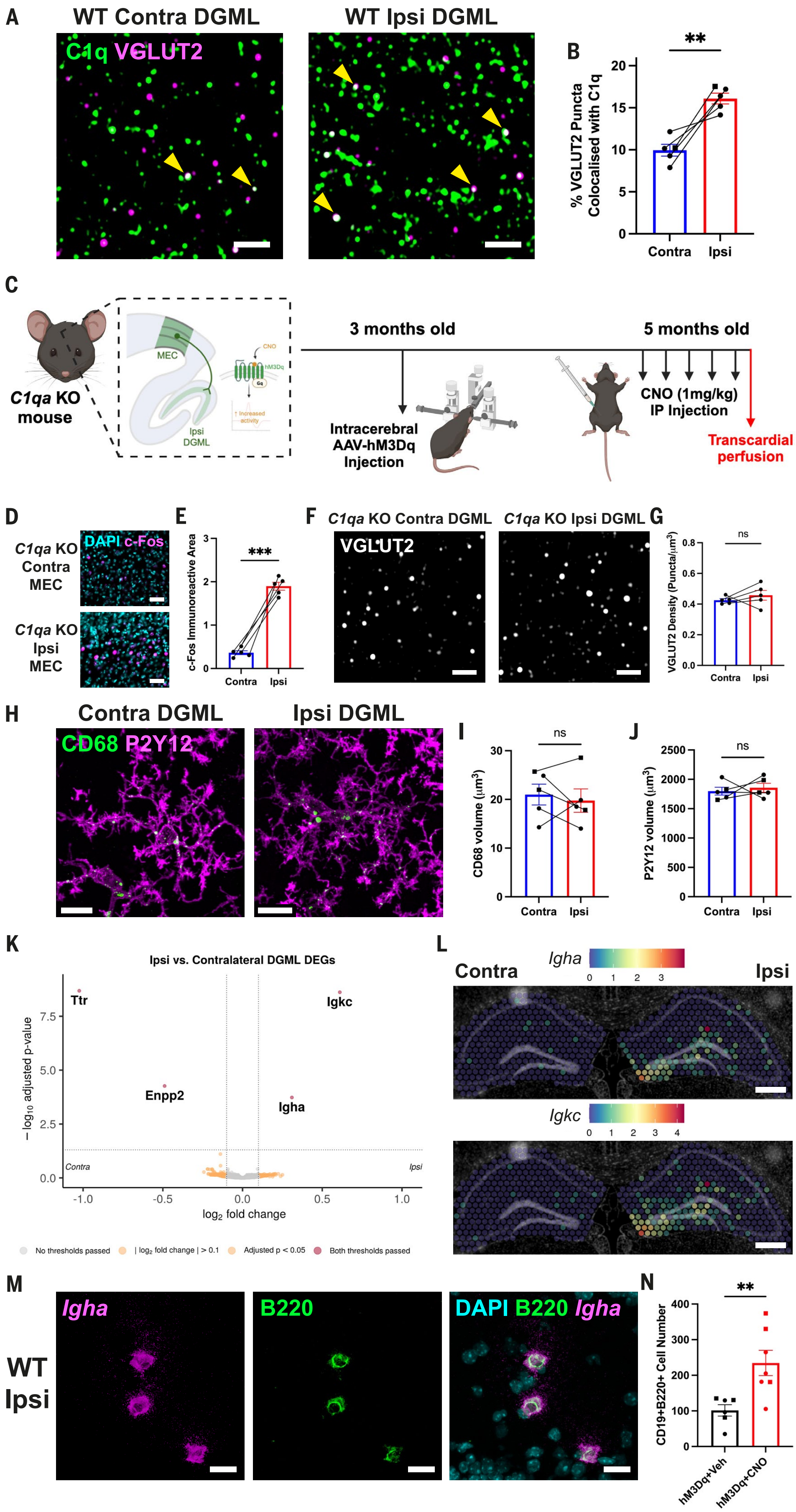

Fig. 3. Activity-dependent synapse loss is accompanied by recruitment of B lymphocyte lineage cells to the hippocampus. (**A**) Representative images of C1q and VGLUT2 colocalization in contralateral and ipsilateral DGML; colocalized puncta are indicated by yellow arrowheads. Scale bars, 2 μm. (**B**) Quantification of percentage total VGLUT2 puncta colocalizing with C1q in contralateral and ipsilateral DGML of hM3Dq+CNO WT mouse. $n = 5$ mice, one brain section per hemisphere, three ROIs analyzed per section. *P* value from paired *t* test. (**C**) Schematic of *C1qa* KO experimental paradigm. Created with BioRender.com. (**D**) Representative widefield images of c-Fos labeling at contralateral and ipsilateral MEC of hM3Dq+CNO *C1qa* KO mouse. Scale bars, 50 μm. (**E**) Quantification of c-Fos–immunoreactive area between hemispheres in hM3Dq+CNO *C1qa* KO mouse. $n = 5$ *C1qa* KO mice, two or three brain sections per hemisphere, three ROIs analyzed per section. *P* values from paired *t* test. (**F**) Representative Airyscan images of VGLUT2 puncta at contralateral and ipsilateral DGML of hM3Dq+CNO *C1qa* KO mouse. Scale bars, 2 μm. (**G**) Quantification of VGLUT2 puncta density between hemispheres in hM3Dq+CNO *C1qa* KO mouse. $n = 5$ *C1qa* KO mice, two to five brain sections per hemisphere, three ROIs analyzed per section. *P* values from paired *t* test. (**H**) Representative images of P2Y12 and CD68 within the contralateral and ipsilateral DGML of the hM3Dq+CNO mouse. Scale bars, 10 μm. (**I**) Quantification of CD68 volume per P2Y12+ microglia analyzed using Imaris. $n =$ 5 hM3Dq+CNO mice, two or three sections per hemisphere, 7 to 12 microglia per hemisphere. *P* values from paired *t* test. (**J**) Quantification of P2Y12 volume analyzed using Imaris. $n = 5$ hM3Dq+CNO mice, two or three sections per hemisphere, 7 to 12 microglia per hemisphere. *P* values from paired *t* test. (**K**) Volcano plot of DEGs at the DGML spot selection. (**L**) Representative images of Visium spot mRNA transcript counts at contralateral and ipsilateral hippocampi for *Igha* and *Igkc*. Scale bars, 500 μm. (**M**) Representative confocal image of B220+ cells expressing *Igha* mRNA at ipsilateral hippocampal DG. Scale bars, 10 μm. (**N**) Quantification of CD19+B220+ cells in hippocampus of hM3Dq+Veh and hM3Dq+CNO WT mice by FACS. $n = 6$ hM3Dq+Veh mice and $n = 7$ hM3Dq+CNO mice. *P* value from unpaired *t* test. Throughout, square points represent males and circular points represent females. Linked points indicate data from the same mouse brain. One point is equal to one mouse or hemisphere average across brain sections. Data are mean ± SEM. $*P < 0.05$, $**P < 0.01$.

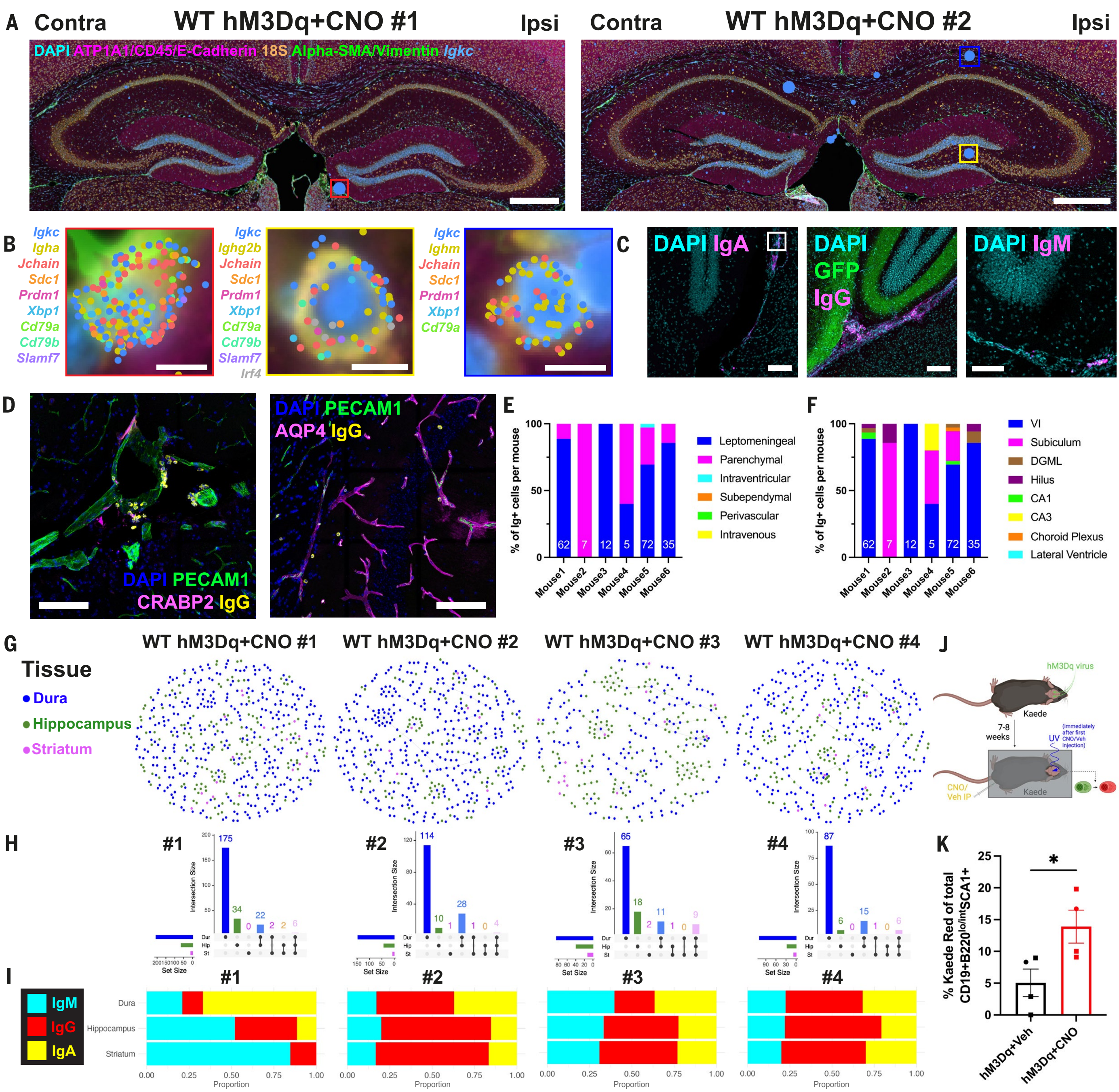

Fig. 4. Hippocampal antibody-secreting cells localize primarily to leptomeningeal areas and share repertoire features with dural B-lineage cells. (**A**) Representative images of Xenium mRNA transcript counts at contralateral and ipsilateral hippocampi for *Igkc* with insets of *Igkc*+ cells in red, yellow, and blue. Scale bars, 500 μm. (**B**) Zoomed insets from Xenium sections in (A) of *Igha*+, *Ighg2b*+, and *Ighm*+ plasma cells. Scale bars, 5 μm. (**C**) Representative confocal images of IgA+ cells (inset represents IgA+ CD138+ cells in fig. S13G), IgG+ cells with activated GFP+ projections, and IgM+ cells at ipsilateral hippocampal DG. Scale bars, 100 μm. (**D**) Representative confocal images of PECAM1, AQP4, and IgG in hippocampal parenchyma and PECAM1, CRABP2, and IgG in the VI. Scale bars, 100 μm. (**E**) Quantification of B cell localization by cellular microenvironment; total number of cells identified per mouse denoted within each bar. (**F**) Quantification of B cell localization by hippocampal subregion; total number of cells identified per mouse denoted within each bar. (**G**) BCR network dandelion plots for separate hM3Dq+CNO mice colored by tissue distribution. Each node corresponds to a single BCR contig, connected nodes indicate that BCR contigs are similar, and distance varies with degree of similarity. (**H**) Upset plots of BCR contig number. Vertical graphs show the number of unique and overlapping BCR contigs for each tissue, with intersections indicated by connected dots below the *x* axis. Number of BCR contigs are indicated above the columns. Horizontal graphs at left show the total number of BCR contigs detected in each tissue respectively. (**I**) Bar graphs demonstrating the proportion of each heavy chain isotype class in the total BCR repertoire across tissues. (**J**) Schematic of Kaede experimental paradigm. Created with BioRender.com. (**K**) Quantification of percentage Kaede Red+ CD19+B220$^{lo/int}$SCA1+ cells at the hippocampus of hM3Dq+Veh and hM3Dq+CNO Kaede mice by FACS. $n = 4$ hM3Dq+Veh mice and $n = 4$ hM3Dq+CNO mice. *P* value from unpaired *t* test. Throughout, circular points represent females and linked points indicate data from the same mouse brain. Data are mean ± SEM. $^*P < 0.05$.

F and G). These data, together with the BCR-seq, suggest that a subset of hippocampal B-lineage cells derives from the skull bone marrow and/or dural meninges in response to local neuronal activity.

Recruitment of B-lineage cells requires chemokine signaling, often occurring via the CXCL12-CXCR4 axis (39, 40). In line, we identified locations where plasma cells are found in our Xenium dataset and tested for differentially expressed genes (DEGs) within each cell type as proximity to the putative plasma cells changes. Within the Capillary Intermediate 1 cell type (includes vascular/leptomeningeal cells), we identified higher expression of *Cxcl12* by cells situated closer to plasma cells compared with those farther away (fig. S17A). In conjunction, within the Late Pro-B cell type (includes plasma cells themselves), we found enriched *Cxcr4* expression close to plasma cell locations (fig. S17B), suggesting potential spatial association of CXCL12-CXCR4 signaling with plasma cell localization; however, this needs to be experimentally validated. Plasma cells also require survival cues such as A proliferation-inducing ligand (APRIL, encoded by *Tnfsf13*) (41–44). We observed increased expression of *Tnfsf13* at the ipsilateral, activated VI compared with the contralateral VI (fig. S18, C to E). *Tnfsf13* expression localized to *Col1a1*+ fibroblasts, primarily at leptomeningeal borders around the hippocampus (fig. S18, F and G), and not to Iba1+ macrophages (fig. S18C). In line, APRIL's receptor B cell maturation antigen (BCMA, encoded by *Tnfrsf17*), which is expressed by mature B cells and plasma cells (41, 44), was detectable in *Igkc*-expressing cells at the ipsilateral VI near the activated DG (fig. S18, A and B). These data suggest that ASCs, likely originating from the skull bone marrow and/or dural meninges, localize to the hippocampus in an activity-dependent manner.

Secreted antigen-specific IgM is required for activity-dependent C1q deposition and synapse loss

To determine the functional role of B lymphocyte lineage cells in the activated hippocampus, we examined whether immunoglobulins participate in the activity-dependent recruitment of C1q to VGLUT2 presynaptic terminals. Although classical complement cascade activation outside the central nervous system (CNS) typically requires C1q fixation to IgG or IgM (45, 46), immunoglobulins are rarely considered within the CNS, likely because of limited presence at homeostasis. Using Airyscan imaging, we detected a higher proportion of C1q+VGLUT2+ terminals colocalizing with IgM-immunoreactive puncta at the ipsilateral DGML compared with the contralateral side (Fig. 5, A and B). In contrast, IgA and IgG puncta were not detected.

This observation is consistent with a role for IgM in facilitating C1q engagement at VGLUT2 presynaptic terminals, analogous to its role in peripheral complement activation (47). However, immunohistochemical colocalization likely underestimates the true extent of transient or weak molecular interactions owing to technical limitations, which can include incomplete epitope detection, subdiffraction spatial organization, and capture of only a momentary snapshot of opsonization events. Given that C1q-mediated tagging and subsequent immune engagement are likely rapid and dynamically regulated to avoid excessive neuroinflammation, low absolute colocalization values are biologically expected. Indeed, prior work suggests a role for C1q in normal hippocampal synaptic remodeling or turnover outside overt disease (48). To further investigate the role of IgM in C1q-mediated synapse loss, we used three mouse models modulating B cell function. First, we tested whether B cells are required for this process by activating the perforant pathway in μMT mice, which lack the IgM heavy chain and thus fail to generate B cells, plasma cells, or secreted immunoglobulins (Fig. 5C) (49). Despite robust hM3Dq-driven activation, evidenced by increased c-Fos in ipsilateral MEC (fig. S19, A and B), μMT mice displayed no activity-dependent differences in C1q deposition (Fig. 5, D and F) or VGLUT2 density (Fig. 5, E and G) between the ipsilateral and contralateral DGML, suggesting that B-lineage cells are required for activity-dependent C1q deposition and synapse loss.

We further assessed the requirement for secretion of soluble immunoglobulins, particularly IgM. To that end, we used secreted IgM (sIgM) KO mice, in which ASCs can be generated (50, 51) but cannot secrete IgM (52) and exhibit impaired IgG responses (Fig. 5H) (53). Other antibody isotypes (IgA, IgG1, IgG2a, IgG2b, IgG3) are present in approximately normal amounts (52). Although hM3Dq+CNO triggered c-Fos up-regulation at the ipsilateral MEC (fig. S19, C and D), sIgM KO mice failed to show the ipsilateral increase in C1q density (Fig. 5, I and K) or reduction in VGLUT2 puncta (Fig. 5, J and L) observed in WT mice (Fig. 1). This suggests that secreted IgM is required for neuronal activity–dependent C1q deposition and synapse loss.

Finally, we tested whether antigen recognition by IgM is necessary for this process using MD4 mice, in which B cells produce monoclonal hen egg lysozyme (HEL)–specific IgM but are unable to generate antibodies to other antigens (54) or to class-switch to IgA or IgG (Fig. 5M) (55). IgM is still secreted by ASCs in these mice (54, 56), therefore MD4 mice specifically test whether the antigen specificity of IgM is relevant for activity-dependent synapse loss. MD4 mice exhibited normal c-Fos up-regulation (fig. S19, E and F) but showed no hemispheric differences in C1q (Fig. 5, N and P) or VGLUT2 density (Fig. 5, O and Q). These findings support a requirement for antigen-specific IgM in the activity-dependent C1q-mediated VGLUT2 loss.

Altogether, these data from μMT, sIgM KO, and MD4 mice suggest that secreted, antigen-specific IgM produced by B-lineage ASCs is required for activity-dependent C1q deposition and synaptic refinement in the adult hippocampus. Other immunoglobulin isotypes may also be involved but are insufficient, as sIgM KO mice do not exhibit changes in C1q and VGLUT2 density, despite the presence of near-normal amounts of IgA and IgG.

We next tested whether the functional relationship between B lymphocyte lineage cells and activity-dependent C1q-synapse phenotypes observed in the chemogenetic models can be applied to other acute models of neuronal hyperactivity. First, we used intraperitoneal administration of kainic acid (KA; 14 mg/kg). As expected (57), KA increased hippocampal c-Fos expression (fig. S20, A and B), confirming elevated neuronal activity. Repeated KA administration for 3 days induced robust C1q deposition in the hippocampal CA3 stratum lucidum (CA3SL; fig. S20, C and D), where c-Fos+ DG granule cell projections terminate, paralleling effects observed in the hM3Dq chemogenetic model (Fig. 1, D and E). In contrast, KA failed to induce C1q deposition in μMT mice (fig. S20, E and F), reinforcing that B lymphocyte lineage cells are required for activity-evoked C1q deposition. However, this 3-day KA paradigm did not produce measurable synapse loss at the CA3SL of WT mice (fig. S20, G and H), likely reflecting insufficient duration or magnitude of hyperactivity.

Considering this along with potential confounds of glutamate receptor overstimulation in KA paradigm, we next used a more refined neuronal hyperactivity model using pentylenetetrazole (PTZ), a γ-aminobutyric acid type A ($GABA_A$) antagonist (58). Adult WT and μMT mice aged 4 to 7 months received PTZ (32 to 40 mg/kg, adjusted for exact mouse age) or saline intraperitoneally twice over 5 days, producing comparable behavioral seizure scores on a modified Racine scale (59). As previously shown (58), PTZ elevated hippocampal DG c-Fos (fig. S21A). However, only WT mice exhibited increased C1q deposition (fig. S21, B and C) and reduced Bassoon+Homer1+ colocalized excitatory synaptic puncta density (fig. S21, D and E). μMT mice showed no PTZ-induced changes in C1q (fig. S21, F and G) or synaptic puncta (fig. S21, H and I). These data indicate that sustained neuronal hyperactivity elicits C1q deposition and structural synapse loss specifically in the presence of B lymphocyte lineage cells.

Finally, we asked whether B-lineage cells could underlie the C1q-Homer1 phenotypes reported in J20 mice (3) (fig. S8, A to D). We did not detect IgM colocalizing with Homer1 and C1q in 1- or 4-month-old J20 mice, nor did we detect B-lineage cells at the hippocampus by immunostaining in J20 mice or in hM4Di+CNO J20 mice (data not

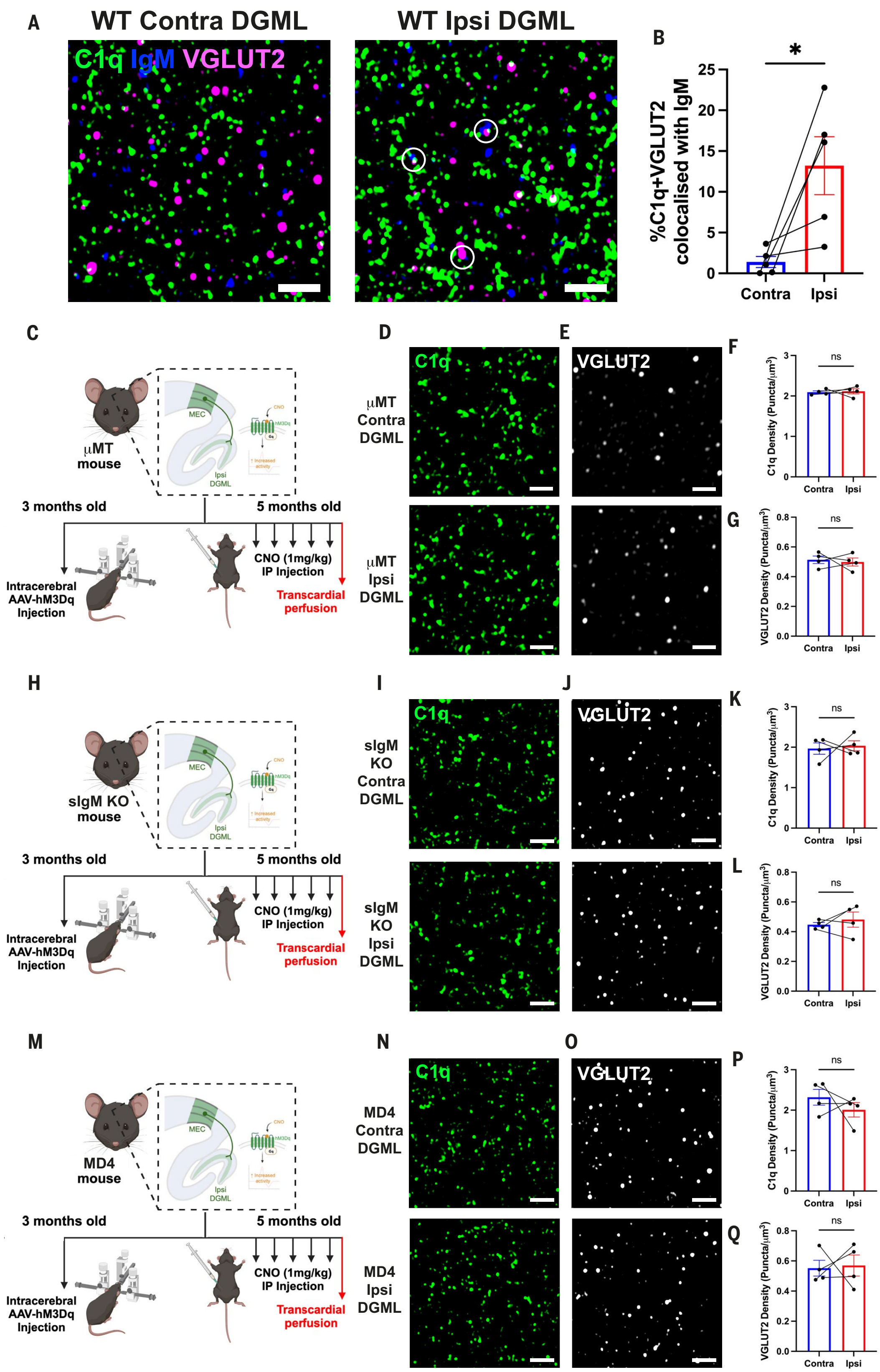

Fig. 5. Immunoglobulins, particularly IgM, are required for activity-dependent C1q deposition and synapse loss. (**A**) Representative Airyscan images of IgM puncta colocalized with double positive C1q+VGLUT2+ puncta at contralateral and ipsilateral DGML of hM3Dq+CNO WT mouse. Scale bars, 2 μm. (**B**) Quantification of colocalized IgM, C1q, and VGLUT2 puncta between hemispheres in hM3Dq+CNO WT mouse. Five hM3Dq+CNO mice, two brain sections per hemisphere, two or three ROIs analyzed per section. *P* values from paired *t* test. (**C**) Schematic of μMT experimental paradigm. Created with BioRender.com. (**D**) Representative Airyscan images of C1q puncta at contralateral and ipsilateral DGML of hM3Dq+CNO μMT mouse. Scale bars, 2 μm. (**E**) Representative Airyscan images of VGLUT2 puncta at contralateral and ipsilateral DGML of hM3Dq+CNO

μMT mouse. Scale bars, 2 μm. (**F**) Quantification of C1q puncta density between hemispheres in hM3Dq+CNO μMT mice. Four μMT mice total, two brain sections per hemisphere, three ROIs analyzed per section. *P* values from paired *t* test. (**G**) Quantification of VGLUT2 puncta density between hemispheres in hM3Dq+CNO μMT mice. Four μMT mice total, two or three brain sections per hemisphere, three ROIs analyzed per section. *P* values from paired *t* test. (**H**) Schematic of sIgM KO experimental paradigm. Created with BioRender.com. (**I**) Representative Airyscan images of C1q puncta at contralateral and ipsilateral DGML of hM3Dq+CNO sIgM KO mouse. Scale bars, 2 μm. (**J**) Representative Airyscan images of VGLUT2 puncta at contralateral and ipsilateral DGML of hM3Dq+CNO sIgM KO mouse. Scale bars, 2 μm. (**K**) Quantification of C1q puncta density between hemispheres in hM3Dq+CNO sIgM KO mice. Four sIgM KO mice total, two brain sections per hemisphere, three ROIs analyzed per section. *P* values from paired *t* test. (**L**) Quantification of VGLUT2 puncta density between hemispheres in hM3Dq+CNO sIgM KO mice. Four sIgM KO mice total, two or three brain sections per hemisphere, three ROIs analyzed per section. *P* values from paired *t* test. (**M**) Schematic of MD4 experimental paradigm. Created with BioRender.com. (**N**) Representative Airyscan images of C1q puncta at contralateral and ipsilateral DGML of hM3Dq+CNO MD4 mouse. Scale bars, 2 μm. (**O**) Representative Airyscan images of VGLUT2 puncta at contralateral and ipsilateral DGML of hM3Dq+CNO MD4 mouse. Scale bars, 2 μm. (**P**) Quantification of C1q puncta density between hemispheres in hM3Dq+CNO MD4 mice. Four MD4 mice total, two brain sections per hemisphere, three ROIs analyzed per section. *P* values from paired *t* test. (**Q**) Quantification of VGLUT2 puncta density between hemispheres in hM3Dq+CNO MD4 mice. Four MD4 mice total, two or three brain sections per hemisphere, three ROIs analyzed per section. *P* values from paired *t* test. Throughout, circular points represent females and linked points indicate data from the same mouse brain. One point is equal to one hemisphere average across brain sections. Data are mean ± SEM. $^{ns}P > 0.05$, $^{*}P < 0.05$.

shown). Given the absence of detectable IgM and B-lineage cell involvement in the pre-plaque J20 brain, we depleted B cells and plasma cells using repeated intravenous administration of anti-CD20 antibody and bortezomib, respectively (*60*), spanning the interval from when synaptic C1q accumulation emerges (1 month old) to when Bassoon+Homer1+ synapse loss becomes detectable (4 months old) (*3*). Flow cytometry confirmed near-complete depletion of both populations from circulating blood at 2.5 and 4 months old (fig. S22A). Despite this, B cell and plasma cell depletion did not alter C1q accumulation (fig. S22, B and C) or excitatory synapse loss (fig. S22, D and E) in J20 mice. Although we cannot rule out the presence of CNS-resident B-lineage cells and immunoglobulins in the pre-plaque J20 brain, these results suggest that, unlike acute activity-driven models, the J20 amyloidopathy model likely exhibits B cell–independent C1q accumulation and synaptic degeneration.

Discussion

Neuronal activity regulates synaptic pruning during development (*61*) and microglia-mediated synapse elimination (*13*), but its role in region-specific C1q deposition and synapse loss in the adult brain has remained unclear. Using in vivo chemogenetics, we show that heightened activity along the perforant pathway leads to focal C1q deposition and synapse loss in the hippocampus. These findings identify neuronal hyperactivity as a physiological trigger for C1q recruitment at adult synapses. Notably, the selective elimination of VGLUT2 presynaptic terminals, while leaving activated VGLUT1 projections intact, resembles competitive activity-dependent refinement seen during development (*62*). The specific molecular cues enabling this selectivity remain undefined. Candidate mechanisms include activity-dependent "eat-me" and "don't-eat-me" signals including PirB (*63*), CD47 (*64*), phosphatidylserine (*65*), local damage-associated molecular patterns (DAMPs) (*66*), and/or glial feedback mechanisms (*67*–*69*).

Our findings thus offer a functional link between neuronal hyperactivity and C1q-mediated synapse loss in the adult CNS, raising the possibility that dysregulated neuronal activity, an early hallmark of AD (*10*, *70*), contributes to region-specific complement deposition and synaptic vulnerability (*3*). In early AD, a pathological feedback loop may arise in which neuronal hyperactivity promotes Aβ and/or tau release at synapses (*21*–*24*), enhancing complement accumulation and increased synaptic susceptibility, potentially through metabotropic glutamate receptor 5 (*71*), while Aβ aggregates further exacerbate hyperexcitability (*70*, *72*). Consistent with this model, reducing perforant pathway activity in 4-month-old J20 mice decreased hippocampal C1q accumulation and partially restored Homer1-immunoreactive synapses. As reduced perforant pathway activity is linked to lower hippocampal Aβ accumulation (*73*, *74*), we propose that in J20 mice, a feedback loop of activity-dependent synaptic Aβ and Aβ-dependent hyperactivity pathologically amplifies complement-mediated synaptic vulnerability (*70*). This aligns with observations that levetiracetam, an antiepileptic drug, alleviates synaptic and cognitive deficits in J20 and APP/PS1 mice (*17*, *75*) and reduces hippocampal hyperactivation while improving hippocampal-dependent memory in individuals with amnestic mild cognitive impairment (*76*). Together, these studies propose neuronal hyperactivity as an early determinant of synapse vulnerability, and that suppressing neuronal hyperactivity may benefit subgroups of individuals with AD exhibiting epileptiform activity (*77*).

More broadly, the contrasting outcomes in WT hM3Dq-activated mice and the J20 hM4Di-inactivated mice underscore that C1q-mediated synaptic refinement is highly context dependent. In WT mice, specific presynaptic terminals (VGLUT2+, not VGLUT1+) were reduced, while postsynaptic components (Homer1+) were unaffected by repeated DREADD activation, whereas in J20 mice, by contrast, DREADD-mediated inhibition affected Homer1+ postsynaptic compartments, in line with previous studies suggesting postsynaptic vulnerability in the hAPP-overexpressing J20 mice (*3*, *78*–*80*). At the 5-day time point of our analysis, the chemogenetic modulation in WT mice preferentially targets presynaptic terminals along the perforant pathway, consistent with activity-dependent, complement-mediated synaptic pruning, in which less-active presynaptic terminals are preferentially eliminated (*81*). These findings suggest that in the healthy adult CNS, presynaptic vulnerability may be governed by local activity differentials, whereas in AD, widespread Aβ-driven pathology appears to induce postsynaptic loss, at least in the circuit examined here.

The unexpected implication from unbiased spatial transcriptomics of local immunoglobulin production in activity-dependent C1q deposition was validated using higher-resolution and orthogonal methods, including single-cell Xenium, RNAScope, and confocal imaging. Further, functional relevance was demonstrated using live-cell tracking, BCR-seq, super-resolution imaging, and chemogenetics in multiple genetic models. These findings altogether suggest that neuronal hyperactivity leads to the accumulation of B lymphocyte lineage cells in the hippocampus at the site of activity-dependent synapse loss in our model. Although there could be confounding factors of surgical damage or viral transduction facilitating B-lineage cell recruitment in the chemogenetic paradigm, control experiments support a primary role for neuronal activity modulation. These include three control mouse groups that underwent identical surgical procedures, yet hemisphere-specific changes were restricted to activity-modulated conditions, and lack of dextran permeability suggesting intact BBB. In addition, independent models of neuronal hyperactivity using KA or PTZ recapitulate key features of the phenotype, supporting a role for neuronal activity independent of surgical manipulation. These B-lineage cells, which normally reside in meningeal immune niches (*36*, *37*), appear to accumulate near the ventricle-hippocampus interface and the VI upon hM3Dq-mediated hyperactivity. Their presence may be induced by the CXCL12-CXCR4 axis and stabilized by APRIL-BCMA signaling, and they produce immunoglobulins, particularly IgM, that are associated with C1q deposition. Using three mechanistically distinct mouse models (μMT, sIgM KO, and MD4), we demonstrate that antigen-specific, secreted IgM from plasmablasts and/or plasma cells is required for activity-dependent C1q deposition and VGLUT2 loss. We further validated in other hyperactivity models the functional relevance of B lymphocyte lineage cells in activity-dependent C1q deposition (KA, PTZ) and synapse loss (PTZ). Although IgM is a well-established activator of C1q in peripheral tissues (*47*), its role in the CNS remains to be fully elucidated. Together, these findings suggest that humoral immune mechanisms can operate within the CNS and facilitate C1q deposition and synapse loss during heightened neuronal activity.

The adaptive immune system's contribution to CNS function and its interactions with microglia and C1q have been poorly defined. Our data support a model in which immunoglobulin-C1q interactions contribute to synapse loss following neuronal hyperactivity. Key mechanistic questions remain: What antigens do these immunoglobulins recognize? Do B cell and plasma cell recruitment and immunoglobulin deposition occur in normal circuit dynamics including development or only during aberrant hyperactivity in the adult brain? Are similar innate-adaptive interactions present in other brain circuits, including those less plastic or farther from immune niches than the hippocampus (*82*)? Natural antibodies, which are immunoglobulins that bind to self-antigens including phosphatidylserine and activate C1 to facilitate removal of apoptotic bodies (*83*), are of particular interest. Moreover, apoptotic signals such as phosphatidylserine are exposed on synapses undergoing elimination (*25*, *65*), raising questions about the presence of natural antibodies in the CNS and their involvement in activity-driven synapse loss. Future work will seek to clarify their role and whether their presence in CNS immune niches is conserved.

Further, whether similar complexes contribute to neurodegenerative disease remains unknown. Proteomic analyses of human AD synapses show IgG3 and C1q enrichment (*84*). In Guillain-Barré models, anti-ganglioside IgM and IgG3 antibodies interact with C1q at neuromuscular junctions to facilitate peripheral neuropathy (*85*), suggesting antibody-dependent C1 activation on synaptic terminals. These observations raise the possibility that immunoglobulin-C1q complexes may also form at CNS synapses in disease. Reports on B cells in AD models have been inconsistent, likely reflecting differences among APP transgenic lines as well as the complexity of adaptive-innate immune cross-talk in chronic diseases (*86*–*93*). In our study, depleting circulating B cells and plasma cells over a 3-month period in young adult J20 mice did not alter C1q deposition or synapse loss. We speculate that the chronic, widespread network hyperactivity and hAPP overexpression in J20 mice yields pathological synapse loss because of the appearance of Aβ-induced DAMPs across multiple brain regions (*1*, *11*), obscuring activity-dependent adaptive immune involvement. By contrast, the WT chemogenetic model produces focal, temporally defined hyperactivity, allowing assessment of localized immunoglobulin-dependent synaptic changes. Although amyloidopathy mouse models have caveats and limited resolution, emerging human data suggest that B cells and immunoglobulins may participate in AD-related vulnerability. A longitudinal study showed circulating B cells and IgG correlate with cerebral amyloidosis and altered microglia states, with common BCR repertoires suggesting convergence on AD-associated antigens of Aβ and tau (*94*). Another study identified a reduction of B cells in the blood of individuals with AD (*95*). Further work in humans and complementary models is needed to resolve the role of B cells and immunoglobulins in AD, including any potential links to neuronal activity.

Our findings demonstrate that neuronal hyperactivity serves as a trigger for C1q deposition and subsequent synapse loss in the adult brain, with immunoglobulin deposition by B lymphocyte lineage cells serving as a key enabling component in the nondiseased brain. We propose an innate-adaptive immune cross-talk mechanism in the CNS that links neuronal activity to complement deposition and synaptic remodeling. Given the relevance of neuronal hyperactivity in neurologic diseases (*96*) and the central role of C1q in synaptopathy (*97*), these findings may have broad implications for understanding synaptic dysfunction across neurologic disorders characterized by circuit impairment. Together, these results position neuroimmune interactions as integral components of circuit regulation, with implications for understanding brain function and vulnerability across physiological and disease states.

Materials and methods

Animals

All animal experiments were performed in accordance with the UK Animals (Scientific Procedures) Act (ASPA) 1986. Experimental procedures were approved by the UK Home Office (Home Office license 70/8999) and followed local ethical advice by consultation with University College London (UCL) veterinary staff. All animal work was carried out under the appropriate ethics and licenses. ARRIVE guidelines were adhered to throughout.

Mice were housed in individually ventilated cages on a 12-hour light/dark cycle in a temperature- and humidity-controlled environment (23.1°C, 30 to 60% humidity) under pathogen-free conditions with ad libitum access to water and food at all times. C57BL/6J WT mice were obtained from Charles River UK. All mouse strains were bred on a C57BL/6J background. hAPP-J20 transgenic mice [J20; B6.Cg-Zbtb20Tg (PDGFB-APPSwInd)20Lms/2Mmjaxn] (*12*) were provided by L. Mucke, University of California, San Francisco, and distributed by P. Salinas, UCL. *C1qa*–/– mice (*C1qa* KO) (*26*) were provided by M. Botto, Imperial College London. Kaede mice were provided by A. Miyawaki, RIKEN Center for Brain Science (*38*). *Ighm*–/– mice (μMT or *Ighm* KO or *Ighm*$^{tm1Cgn/tm1Cgn}$) (*49*) were obtained from JAX (C57BL/6.129S2-Ighmtm1Cgn/J, strain #002288) via the Francis Crick Institute central mouse repository. sIgM KO (*Ighm*$^{tm1Che/tm1Che}$) mice were provided by C. Binder, Medical University of Vienna (*52*). MD4 mice (Tg(IghelMD4)4Ccg) (*54*) and Aicda-CreERT2/R26ReYFP [*Aicda*$^{tm1.1(cre/ERT2)Crey}$ and Gt(ROSA)26Sor$^{tm1(EYFP)Cos}$] mice (*98*) were bred at the Francis Crick Institute.

For all experiments, appropriate sex- and age-matched controls were used. Both male and female mice were used throughout the experiments presented, and details of mouse sex are included in graphs and figure legends. Pilot experiments were conducted using both female and male mice to ensure there were no sex-specific phenotypes. Mice were housed individually for at least 1 month after surgical procedures to avoid compromising sutures. Genotyping for all mouse lines was performed by Transnetyx using tissue from ear punch samples taken to identify mice. For all experiments, the appropriate genotype-, sex-, and age-matched controls were used.

For transcardial perfusion and tissue harvest, mice were deeply anesthetized by intraperitoneal injection of a sodium pentobarbital (200 μg/g body weight; Dolethal, Vetoquinol) overdose. Upon loss of hindlimb and tail pinch reflexes, the abdomen was opened and the heart exposed. A needle was inserted into the left ventricle of the heart and the right atrium punctured using scissors. Twenty milliliters ice-cold filtered phosphate-buffered saline (PBS; 18912, Thermo Fisher Scientific) was pumped at a steady rate through the circulation followed by 20 ml ice-cold ultrapure 4% paraformaldehyde (PFA; 18814-20, Generon; diluted in 0.2 μm-filtered PBS). After perfusion, the mouse head was removed and the brain carefully extracted from the skull followed by postfixation for 24 hours in 4% PFA at 4°C.

Stereotaxic surgery

Stereotaxic surgeries were performed as previously described (*99*, *100*). Briefly, adult mice (3 months old for WT hM3Dq experiments, 2 months old for J20 hM4Di experiments) were anesthetized with isoflurane (IsoFlo, Zoetis) at 4% in oxygen for induction. Mice were head-fixed into a digital stereotaxic frame (504926, World Precision Instruments), and anesthesia was maintained at 1.5 to 2% isoflurane concentration in 1.5 liters/min oxygen flow. Aseptic conditions were adopted throughout. Sterile liquid gel drops (Viscotears, Bausch and Lomb) were applied to keep the eyes of anesthetized mice moist. After cleaning the scalp with a 1:1 solution of 0.9% saline (Aqupharm 3, Animalcare):chlorhexidine gluconate (4% w/v Hibiscrub, Regent Medical), bupivacaine local anesthesia (0.025% Marcain Polyamp, AstraZeneca) was applied and a single skin incision along the scalp midline exposed the skull. Bregma was identified and the skull leveled in the mediolateral (ML) and anteroposterior (AP) aspects. Sterile 0.9% saline was applied at regular intervals to prevent the skull and connective tissue from excessive drying. A craniotomy was drilled using an OmniDrill35 Micro Drill (503599, World Precision Instruments) with 0.9 mm burr (#330104001001007, Hager and Meisinger) unilaterally (always right hemisphere) at stereotaxic coordinates AP −4.60, ML 3.00, dorsoventral (DV) −3.00 (all in millimeters; from Paxinos and Franklin's The Mouse Brain in Stereotaxic Coordinates, Fourth Edition) relative to bregma to target the MEC. To target the CA1 pyramidal layer, coordinates used were AP −2.15, ML 1.50, and DV −1.50. Pulled long-shaft borosilicate pipettes (3.5-inch tubes, #3-000-203-G/X, Drummond Scientific) were backfilled with mineral oil (330779, Sigma-Aldrich) before being loaded with either DREADD virus [AAV8-CaMKIIa-HA-hM3Dq-IRES-mCitrine, #50466-AAV8, Addgene; titer = 1.6×10^{13} genome copies (gc)/ml or AAV8-CaMKIIa-HA-hM4Di-IRES-mCitrine, #50467-AAV8, Addgene; titer = 1.6×10^{13} gc/ml] or control EGFP virus (AAV8-CaMKIIa-EGFP, #50469-AAV8, Addgene; titer = 2.3×10^{13} gc/ml). Controlled injection of 200 nl virus at 80 nl/min rate was conducted using a NanoFil 10 μl syringe (World Precision Instruments) connected to an UltraMicroPump-3 (World Precision Instruments) via NanoFil injection holder (World Precision Instruments) and SilFlex tubing (World Precision Instruments). The pipette was left in place for 10 min after injection, followed by slow retraction to prevent the backflow of contents. The scalp incision was sutured using coated Vicryl polygalactin 910 undyed braided absorbable sutures with 11 mm 3/8c reverse cutting Ethalloy Prime hook (W9500, Ethicon) and closed with Vetbond tissue adhesive (1469SB, 3M) to seal the scalp incision. After incision closure, mice were allowed to recover postsurgery for 1 hour in a heated chamber. Subcutaneous carprofen (5 μg/g body weight; Carprieve, Norbrook) and buprenorphine (0.1 μg/g body weight; Vetergesic, Ceva) diluted in sterile 0.9% saline were administered perioperatively, and carprofen was added to the drinking water (33.33 μg/ml) for 3 days after surgery.

Acute head-fixed recordings using Neuropixels 1.0 probes: Headplate and craniotomy surgery

Stereotactic injection of AAV into the MEC was conducted when WT mice were 3 months old (AAV-CamKII-hM3Dq) and J20 mice were 2 months old. Following full recovery from their first surgery, mice underwent a second surgery to permit acute head-fixed recordings with Neuropixels 1.0 probes.

Anesthesia was induced at 4% isoflurane and maintained at 1.5 to 2%. Mice were subcutaneously injected with 0.1 mg/kg buprenorphine, 0.6 mg/kg meloxicam, and 2 mg/kg dexamethasone. The fur on the scalp was shaved, and betaiodine and lidocaine were applied. The skin was excised to expose the dorsal surface of the skull, and the periosteum and overlying connective tissue were removed and the bones were scored. A ~1-mm-diameter craniotomy was drilled out over the right MEC injection site, which was visible on the skull at AP −4.60, ML 3.00 from bregma (R&S Diamond Bur Taper Conical End Super Fine, Z12; NSK contra angle handpiece, Volvere i7). After removal of the encircled bone at the center of the craniotomy, the dura was immediately moistened with sterile saline (0.9%), and any bleeding was stemmed with sterile hemostatic sponges (Hygitech). A 0.5-mm-diameter silver wire (265586, Sigma) that had been soldered into the female end of a gold-coated header pin (283-5501, RS components) served as a ground during recordings—it was carefully lowered into a ~0.7-mm burr hole, located over the contralateral hemisphere (AP −4, ML −2) so that it touched the dura. Blue-light-curing cement (3M RelyX Unicem 2 Automix) was used to affix the ground pin to the skull and form a small well around the craniotomy, which was then covered with a small hemosponge soaked in silicone oil (378402, Sigma) and sterile saline. Medbond (Animus) glue was applied around the skin perimeter to affix the skin to the skull. A custom stainless steel headplate was centered ~1 mm posterior to bregma using a 3D printed headplate holder. The headplate and skull were covered with Super-Bond (Sun Medical) dental cement. The craniotomy was protected by covering the soaked hemosponge with low-viscosity silicone sealant (Smooth-On). Once the silicone cured, thin strips of blue-light-curing cement were applied over the top of the silicone to secure it.

Habituation and intraperitoneal cannulation

After recovery from the headplate surgery, mice were progressively habituated to head-fixation and the experimenter. On the day of recording, animals were weighed and 0.8 g was deducted to account for the weight of the headplate and cement when calculating the appropriate dilution of CNO dihydrochloride (5 mg/ml; HB6149, Hello Bio). A sufficient volume of 5 mg/ml CNO solution was diluted in 0.9% saline to generate a concentration for a dose of 1 mg CNO/kg body weight with a 100 μl volume injection. Diluted CNO was kept on ice and shielded from light and used within 3 hours of preparation.

The intraperitoneal space of the mice was cannulated to allow for substance administration so as to record changes in MEC unit activity produced CNO-mediated DREADD activation. Immediately before electrophysiology recording, the mouse was anesthetized with isoflurane, induced at 4% and maintained at 0.5 and 1%, and the skin overlying the left side of the peritoneum was shaved and topical lidocaine was applied. The mouse was placed on a heated blanket with its head inside a 10 ml syringe that was continuously supplied with isoflurane. A beveled 18G needle was inserted through both the skin and peritoneal membrane. Flexible polyethylene tubing (BTPE-10, 0.6 mm outer diameter, Instech) that was filled with sterile saline was fed through the needle into the intraperitoneal space. The 18G needle was withdrawn and the polyethylene tube was secured in place by applying topical tissue adhesive (GLUture) to the tube, needle incision locus, and surrounding skin. The dead volume of the cannula was measured to be 30 μl and accounted for when administering CNO during the recording. Once the skin adhesive was cured (~5 min), the animal was allowed to recover from anesthesia and head-fixed into a custom head-fixation apparatus, which was located inside a Faraday cage. The protective silicone that covered the craniotomy was removed, and the dura was immediately hydrated with saline.

Electrophysiological recordings

Neuropixels 1.0 probes (IMEC), which have a four-column checkerboard pattern of recording sites [pitch: 16 μm (column), 20 μm (row)], were sharpened at the tip using a microgrinder (EG-400, Narishige). Before each recording, the probe shank was cleaned with tergazyme [1%, in double-distilled water (ddH_20), Alconox], rinsed in ddH_20, and sterilized with isopropyl alcohol, and the tip of the shank was dipped in DiD (V22887, Invitrogen) to enable histological probe reconstruction.

Probes were connected to an IMEC headstage, and data acquisition was controlled by SpikeGLX software (Version=20231207, https://billkarsh.github.io/SpikeGLX/), which interfaced with a PCIe-8381 card connected to a PXIe-1071 PXI ExpressChassis containing a Neuropixels PXIe-1000 Acquisition Module. Neural data was amplified, multiplexed, filtered, and digitized into a separate AP band (gain = 500, bandwidth = 0.3 to 10 kHz, sampling rate = 30 kHz) and LFP band (gain = 50, bandwidth = 0.5 to 500 Hz, sampling rate = 2.5 kHz). Probe ground and reference pads were shorted by soldering them to a silver wire, which was, in turn, connected to the ground pin on the skull of the mouse during recordings.

Probes were affixed to a four-axis micromanipulator (Sensapex, uMp-4) by way of a MPM-Probe Mount-1AS (NewScale Technologies). The angle of insertion of the probe shank was perpendicular to the headplate/skull (i.e., a vertical insertion). The tip of the shank and craniotomy site were visualized with a trinocular microscope. Typically, the insertion site of the previously conducted virus injection was visible on the dura. The shank tip was lowered until it made contact with the dura, the vertical manipulator axis was zeroed, and then the shank inserted through the brain at a speed of 2 to 5 μm/s until it reached a final depth of 3 mm and allowed to settle for ~10 min.

The recording was acquired as one continuous session but had three stages: (i) at 20 min, 50 μl of saline was delivered through the cannula; (ii) at 50 min, 100 μl of CNO was delivered through the cannula; (iii) a further 90 min of continuous recording post-CNO administration. After the recording finished, mice were administered sodium pentobarbital through the cannula. After loss of consciousness and loss of hindlimb and tail pinch reflexes, the abdomen was opened and transcardial perfusion and brain fixation was performed as outlined earlier. In all six mice, proper cannulation of the intraperitoneal space was confirmed upon opening of the abdomen.

Preprocessing, spike sorting, and histological alignment with GFP intensity

Recording session ap.bin files were preprocessed using the local common average reference function in CatGT (https://billkarsh.github.io/SpikeGLX/) and then spike-sorted between a start time of 1500 s until the end of the recording using Kilosort 4 (version = 4.0.37), with default parameters drift correction enabled (nblocks = 5) (*101*).

Reconstruction the trajectory of the shank was achieved using serial 30 μm sagittal brain slices spanning the MEC with anti-GFP antibody and confirmatory anti-c-Fos or anti-pPDH antibody. Slices were aligned to the Allen CCF mouse atlas and probe trajectories delineated using APhistology (https://github.com/petersaj/AP_histology). The trajectory output from APhistology was then loaded into UniversalProbeFinder (https://github.com/JorritMontijn/UniversalProbeFinder) to permit final alignment of the histology with the spatially localized spiking data from Kilosort4, as well as the maximum channel in the brain that was estimated from each recording's lf.bin and ap.bin file. In one animal (WT#1), the shank passed through the visual cortex and MEC and then pierced out the posterior surface of the MEC and entered the cerebellum. This was evident in the DiD deposit and the spatially localized spiking data.

In all six animals, the probe trajectory intersected regions of the MEC containing brightly labeled GFP-positive, DREADD-expressing neurons. However, the position and extent on the GFP expression in relation to the probe shank varied: In some recordings (e.g., J20#1), GFP expression was strong only toward the shank's tip, in other recordings (e.g., WT#2), GFP-positive expression spanned large portions of the shank. We used an image-based thresholding approach to estimate parts which parts of the shank and, in turn, recording channels, that were located in strongly GFP-expressing areas. For each recording, the slice image in which the tip of the probe was visible on the DiD channel was opened in FIJI and a rectangular region of interest (ROI) was positioned at the probe's most ventral tip and, following the trajectory of the probe dorsally, extended until at the MEC/V1 border. The dimensions of the ROI were 100 μm wide (to approximate a 70 μm wide Neuropixels 1.0 probe), and 1600 to 1800 μm long (typical distance between the tip of the shank and the MEC-V1 border). After switching to the GFP channel, we used this ROI in combination with the "Plot Profile" analysis tool to measure the raw GFP fluorescence intensity values, which were extracted as a function of distance from the tip of the shank.

To account for background fluorescence and permit binary thresholding, for each image, we obtained the GFP fluorescence intensity values using 200 μm by 200 μm ROI positioned region of the slice with no visible GFP-positive neurons. Binary thresholding (GFP-positive or GFP-negative) was achieved as follows

$$\text{Threshold} = \mu_{\text{background}} + (k \cdot \sigma_{\text{baseline}})$$

where $\mu_{\text{background}}$ and σ_{baseline} are the mean and standard deviation of the 200 μm by 200 μm background ROI and k is a multiplication factor that was set to 10.

Next, the fluorescence intensity profiles along the probe trajectory were spatially binned by taking mean fluorescence in consecutive 20 μm bins, which corresponds to the row separation between pairs of channels on the Neuropixels 1.0 probe. Twenty-micrometer bins exceeding the calculated threshold and the putative associated channels were assigned as "GFP-positive," while those bins and putative associated channels below the threshold were assigned "GFP negative."

CNO-induced changes to unit activity

The changes in activity induced by CNO-mediated DREADD activation were assessed using custom MATLAB scripts (2024a). Functions in the Spikes toolbox (https://github.com/cortex-lab/spikes) were used to (i) import the spike-sorting results from Kilosort 4, from which we only used units classified by Kilosort 4 as "good" single units for analysis; and (ii) assign these good units to the channel with the largest average spike amplitude. Using this, combined with the binary thresholding approach discussed above, allowed us to separate good units into those putatively associated with high GFP expression: "GFP-positive units"; and those that are not: "GFP-negative units." Comparisons in CNO-induced changes to neural activity could then be assessed in "GFP-positive units" across time (e.g., before and after CNO), and between groups (e.g., the change in activity of "GFP-positive units" relative to "GFP-negative units").

All analysis was performed by dividing the recording into 10-s bins—six of these bins were averaged for the purpose of graphical visualization across time. The effect of CNO was always quantified relative to the baseline period, which was defined as the 600 s prior to CNO administration (60× 10-s time bins from 40 min into the recording to the 50 min time point, when CNO was administered). For good single units to be included in the analysis, they had to have a mean firing rate exceeding 0.1 Hz during the baseline period.

We used two metrics to quantify and graphically illustrate changes in neural activity: (i) changes to mean population firing rate (MFR), and(ii) mean z-scored change in firing rate. MFR is simply the average spike rate of all good units in each time bin. Mean z-score was calculated on a unit basis, whereby the mean and standard deviation of the firing rate across the 60× 10-s baselines bins was calculated. The z-score change in firing within each subsequent 10-s time bin was calculated by, first, subtraction of the mean baseline firing rate and then dividing by the standard deviation of the firing rate across the baseline bins. For statistical analysis and summarizing the overall change in activity induced by CNO, we compared the baseline 600 s of activity to the 600 s at the end of the recording, 80 to 90 min after CNO administration.

Neuropixels statistical analysis

Two statistical comparisons were made. First, differences in MFR for "GFP-positive units" during the baseline period compared with the last 600 s were assessed using a linear mixed-effects model (LMM), with variability due to random effects (animal and cell) taken into account, to allow for a direct assessment of the effect of CNO-mediated DREADD activation. Second, we compared the mean z-scored firing rate change induced by CNO between putative "GFP-positive units" versus "GFP-negative units" during the last 600 s. Mixed-effects models were fitted using Lme4 R package (*102*). For the first comparison, time (e.g., before versus after CNO) was set as a fixed-effect parameter and animal and cell set as nested random-effect parameters. For the second comparison, group was set as the fixed-effect parameter and animal was set as a random-effect parameter. Significance of fixed effects was evaluated using F tests with Satterthwaite-approximate degrees of freedom.

UV photoconversion

Kaede mice undergoing UV photoconversion underwent stereotaxic surgery as described. After scalp incision, the exposed skull was illuminated with a UV lamp at ~100 mW power held 5 mm above lambda for five periods of 1 min each, with a 1-min break period between each illumination cycle. During break periods, saline was applied to keep the skull area cool and moist. The rest of the mouse was covered with aluminum foil during UV illumination to prevent undesired photoconversion.

Substance administration

Kainic acid (KA; 14 mg/kg body weight; K0250, Sigma-Aldrich) freshly dissolved in sterile 0.9% saline or equal volume 0.9% saline vehicle were administered intraperitoneally daily on three consecutive days followed by transcardial perfusion 90 min after the final injection. For 90 min after KA injections, mice were monitored over 30 min for seizure-like activity and scored on a modified Racine scale (table S1) (*103*, *104*). For DREADD experiments, intraperitoneal CNO administration began a minimum of 7 weeks after AAV delivery. A 5 mg/ml working solution of CNO dihydrochloride (water-soluble; HB6149, Hello Bio) dissolved in 0.9% saline was stored at −80°C until aliquoting, and 100 μl aliquots were stored at −20°C until day of use. Sufficient volume of 5 mg/ml CNO solution was diluted in 0.9% saline after weighing mice just before the first intraperitoneal dose to generate a concentration allowing a dose of 1 mg CNO/kg body weight with a 100 μl volume injection. One hundred microliters diluted CNO solution or 100 μl 0.9% saline vehicle was administered IP daily on five consecutive days. Mice were transcardially perfused 90 min after the final CNO or vehicle injection. Mice administered CNO were not observed to display seizure-like activity as measured on the modified Racine scale. For PTZ experiments, PTZ (32 to 40 mg/kg body weight; 2687, Tocris Bioscience) was prepared fresh in sterile 0.9% saline. Subthreshold doses were delivered by intraperitoneal injection, and saline control mice received volume-matched injections. On days of injection, mice were weighed and single-housed into clean, individually ventilated cages; enrichment and food hoppers were removed to prevent impact injury during convulsions. Seizure behavior was observed continuously for 30 min, scored according to the modified seven-point Racine scale (*59*). Diazepam (5 to 10 mg/kg) was administered if mice displayed prolonged stage 6 (>15 s). Mice were monitored for another 60 min and returned to their home cage when exploratory locomotion and grooming returned. To sustain network hyperexcitability, a second subthreshold PTZ injection was administered 48 hours after the first intraperitoneal dose. Saline animals underwent identical handling and timing. Mice were transcardially perfused 48 hours after the second dose. All mice were cohoused up until the day of the experiment. Mice administered with diazepam were not included in the analysis. Behavioral scores: 1, sudden behavior arrest; 2, facial jerking; 3, neck jerks; 4, clonic seizure or sitting seizure; 5, tonic-clonic seizures (lying on belly); 6, tonic-clonic seizure (lying on side) or wild jumping; 7, tonic extension, possibly leading to respiratory arrest and death (*59*). For tamoxifen injections into Aicda-CreERT2/R26ReYFP mice, 100 μl of a 20 mg/ml freshly prepared stock solution of tamoxifen (2 mg dose) dissolved in corn oil was administered immediately before CNO injection on five consecutive days. For dextran experiments, all of 10 kDa dextran-tetramethylrhodamine (Invitrogen, D1868), 70 kDa dextran–Texas red (Invitrogen, D1864), and 2000 kDa dextran-fluorescein (Invitrogen, D7137) were combined at 10 mg/ml each to generate a dextran solution, of which 100 μl was injected intravenously via the tail vein into WT hM3Dq+Veh and hM3Dq+CNO mice. Mice were sacrificed by cervical dislocation 2 min after intravenous injection, followed by brain harvest, freezing in isopentane, cryosectioning, and IHC on the same day to immediately capture any BBB leakiness. To deplete B cells from the circulation over a 3-month period, WT and J20 mice aged 1–4 months old were administered intravenous injections at the tail vein of bortezomib weekly (0.75 mg/kg body weight; 7282, Tocris Bioscience) and anti-CD20 antibody monthly (10 mg/kg body weight; 152115, BioLegend).

Immunohistochemistry

Brains were washed thoroughly in PBS for at least 2 hours at room temperature (RT) followed by incubation for at least 2 days in 30% sucrose (S0389, Sigma-Aldrich) solution at 4°C. After sinking in sucrose solution, brains were embedded in optimal cutting temperature (OCT) compound (4583, Tissue-Tek) and frozen in molds using 2,3-methylbutane (M32631, Sigma-Aldrich) on dry ice. Horizontal and sagittal sections were cut on a cryostat. Fifteen-micrometer-thick sagittal sections (0.3 to 0.6 μm in mediolateral plane) were placed directly onto Superfrost Plus GOLD glass slides (K5800AMNZ72, Thermo Fisher Scientific) for some IHC and all RNAscope experiments while 30-μm-thick free-floating sagittal (0.6 to 1.5 μm in mediolateral plane) and horizontal sections (2.1 to 3.3 μm in dorsoventral plane) were placed in wells of 24-well plates with 1.5 ml cryoprotectant solution [40% PBS, 30% ethylene glycol (324558, Sigma-Aldrich), 30% glycerol (G7757, Sigma-Aldrich)] using stainless steel forceps. Ten sections were stored in each well, meaning each well spanned 0.3 mm in each plane of sectioning, ML for sagittal sections and DV for horizontal sections. Sections on glass slides were stored at −80°C, and free-floating sections were stored in cryoprotectant at −20°C until IHC.

Free-floating sections were left shaking on an orbital shaker at ~90 rpm during all washing, blocking, and antibody incubation steps. Washing consisted of a minimum of three 10-min washing steps in PBS. If a primary mouse antibody was used, M.O.M. (Mouse on Mouse) Blocking Reagent (Vectorlabs, MKB-2213-1) was added before blocking. Sections for immunostaining were first washed then blocked and permeabilized for 2 hours at RT in 20% normal goat serum (NGS; ab7481, Abcam) [20% normal donkey serum (NDS; ab7475, Abcam) used instead when IgA included] and 1% Triton X-100 (28314, Thermo Fisher Scientific) diluted in PBS. Primary antibodies were incubated overnight at 4°C in 10% NGS and 0.5% Triton X-100 in PBS. Washing was followed by secondary antibody incubation in PBS with 10% NGS and 0.5% Triton X-100 for 2 hours at RT. Sections were again washed before 1:10000 4′,6-diamidino-2-phenylindole (DAPI; D1306, Thermo Fisher Scientific) staining for 10 min before mounting.

Sections for synapse marker (Bassoon+Homer1) immunostaining were first washed then pretreated with 1% Triton X-100 before blocking. These sections were then blocked and permeabilized for 2 hours at RT in 20% NGS, 1% bovine serum albumin (BSA; A2153, Sigma-Aldrich), and 0.3% Triton X-100 in PBS. Primary antibodies were incubated overnight at 4°C in 20% NGS, 1% BSA, and 0.3% Triton X-100 in PBS. Wash steps preceded secondary antibody application at 1:500 in 20% NGS, 1% BSA, and 0.3% Triton X-100 for 4 hours at RT. Washing was followed by 1:10000 DAPI staining for 10 min before a final wash and mounting.

For c-Fos immunostaining, sections were washed then blocked for 1 hour at RT in 10% horse serum (26050-070, Thermo Fisher Scientific) and 1% Triton X-100 in PBS. Primary and secondary antibodies were also incubated in 10% horse serum and 1% Triton X-100 in PBS at 4°C overnight and at RT for 2 hours, respectively. Washing steps were as in previous protocols and 1:10000 DAPI was applied for 10 min before mounting. All slides were mounted on Superfrost glass slides (ISO 8037/1, Epredia) using Prolong Gold Antifade mountant (P36930, Thermo Fisher Scientific) with 22 mm by 50 mm #1.5 coverslips (15797582, Menzel-Glaser).

In situ hybridization (RNAscope)

RNAscope ISH experiments were performed as per the manufacturer's specifications using the RNAscope Fluorescent Multiplex Assay (320293, Advanced Cell Diagnostics Bio) in an RNase-free environment. Briefly, 15 μm brain sections on Superfrost Plus GOLD slides (Thermo Fisher Scientific, K5800AMNZ72) were first incubated in hydrogen peroxide (1.07209, Sigma-Aldrich) for 4 min at RT and then washed in RNase-free water. Slides underwent target antigen retrieval at boiling point for 4 minutes and dehydration in 100% ethanol for

5 minutes followed by Protease Plus treatment for 15 min at RT before probes against the target gene (*C1qa*: 441221-C2, *Igha*: 414281-C2, *Igkc*: 414291-C2, *Tnfsf13*: 547331-C1, *Tnfrsf17*: 414871-C1; all Advanced Cell Diagnostics Bio) hybridization for 2 hours at 40°C. Alternating amplification and wash steps followed before addition of Opal 570 fluorophore. Slides were then processed using the IHC protocol above for staining of P2Y12 microglia marker or B220 B cell marker and mounting of slides.

For *C1qa* mRNA analysis, maximum intensity projections were generated on ImageJ before conversion to .ims files. For *C1qa* counting within P2Y12+ cells, number of *C1qa* puncta within P2Y12 channel area of maximum intensity projections was calculated.

FACS experiments

To obtain single-cell suspensions from hippocampus, brains were first quickly isolated from the skull and placed on ice. The dorsal part of the skull was carefully removed, meningeal dura mater was peeled off the skull cap, and hippocampus was dissected from the rest of the brain tissue on ice using chilled instruments. The tissues were finely chopped using chilled razorblades and transferred to tubes containing ice-cold RPMI-1640 supplemented with 10% FBS and 20 mM HEPES medium. Single-cell suspensions from brain tissue were prepared using the Dissociation kit (Miltenyi Biotec) according to the manufacturer's instructions. Tissue chunks were pelleted by centrifugation at 300*g* for 2 min at 4°C followed by medium removal and resuspension in a mix of buffer Z with enzymes A, P, and Y. For blood flow cytometry, circulating blood was extracted from the saphenous vein into heparin-coated tubes and 50 μl was processed. Samples were spun at 300*g* for 5 min and the pellet was resuspended in 1 ml of Red Blood Cell Lysis Buffer 1X (Thermo Fisher Scientific, 00-4333-57) for 10 min to lyse red blood cells. Cells were then washed and spun again for subsequent antibody staining. Single-cell suspensions were then blocked with rat anti-mouse CD16/CD32 antibodies (BD Biosciences) used at the recommended dilution for 12 min before incubation with primary antibodies diluted at 1/200 in fluorescence-activated cell sorting (FACS) buffer (PBS, 2% FBS, 0.78 mM EDTA) containing Fc block for 20 min at 4°C. Dead cells were excluded using Live/Dead near-IR dye staining (LIVE/DEAD Fixable Near-IR Dead Cell Stain; Invitrogen, L34975) diluted in PBS. PE channel was used for detection of Kaede-Red, and Alexa Fluor 488 channel was used for detection of Kaede-Green. Fifty microliters of CountBright Absolute Counting Beads (Invitrogen, C36950) was added to the final sample to measure absolute cell numbers. Flow cytometry data were analyzed using FACSDiva software (v4.0) and FlowJo software (Treestar).

Antibodies

For mouse immunostaining, the following primary antibodies were used: mouse anti-human/mouse 4G8 (BioLegend, 800708; 1/500), mouse anti-human 6E10 (BioLegend, 803001; 1/500), rat anti-mouse B220-AF647 (BioLegend, 103226; clone RA3-6B2, 1/100), rabbit anti-mouse AQP4 (Abcam, ab282586; 1/500), rabbit anti-mouse Bassoon (Synaptic Systems, 141003; 1/200), rabbit anti-mouse c-Fos (Synaptic Systems, 226003 and 226008; 1/500), rabbit anti-mouse C1q (Abcam, ab182451; 1/200), rat anti-mouse CD11b (Bio-Rad, MCA711G; 1/200), rat anti-mouse CD68 (Bio-Rad, MCA1957; clone FA-11, 1/200), rat anti-mouse CD138 (BD Biosciences, 553712; 1/100), rabbit anti-mouse CRABP2 (Proteintech, 10225-1; 1/100), chicken anti-GFP (Abcam, ab13970; 1/500), rabbit anti-mouse GLUT1 (Millipore Sigma, 07-1401; 1/5000), rat anti-HA (Roche, 12158167001; clone 3F10, 1/200), chicken anti-mouse Homer1 (Synaptic Systems, 160006; 1/200), rabbit anti-mouse Iba1 (Wako Chemicals, 019-19741; 1/500), goat anti-mouse IgA (Thermo Fisher Scientific, 62-6700, 1/200), rat anti-mouse IgM (Invitrogen eBioscience, 14-5790-82, 1/200), rabbit anti-mouse P2Y12 (Anaspec, AS-55043A; 1/500), rat anti-mouse PECAM1/CD31 (BD Pharmingen, 550274; 1/200), rabbit anti-mouse phospho-pyruvate dehydrogenase α1 (Cell Signaling Technology, 37115; 1/500), guinea pig anti-mouse VGLUT1 (Millipore Sigma, AB5905; 1/1000), chicken anti-mouse VGLUT1 (Synaptic Systems, 135316; 1/1000), and guinea pig anti-mouse VGLUT2 (Millipore Sigma, AB2251-I; 1/1000).

Secondary antibodies used were a combination of Alexa Fluor 488, 594, and 647 (1/200, Thermo Fisher Scientific) from either donkey or goat where appropriate. For IgG staining, goat anti-mouse F(ab′)2 AF594 (Thermo Fisher Scientific, A11020) was used.

For flow cytometry, the following antibodies were used: CD16/CD32 [Fc Block; BD Biosciences, clone 2.4G2 (RUO)], CD45-BUV395 (BD Biosciences, clone 30-F11, 563792), CX3CR1-BV421 (BD Biosciences, clone Z8-50, 567531), CD3-BV421 (BD Biosciences, clone 145-2C11, 562600), B220-APC (BD Biosciences, 553092), CD19-BV711 (BioLegend, 115555), and SCA1-PeCy7 (BD Biosciences, 561021).

Visium spatial transcriptomics

The Visium Gateway Package from 10x Genomics (PN-1000316, 10x Genomics) was used to generate spatial transcriptomic data from two coronal brain sections from two different hM3Dq-expressing CNO-treated mice. Experiments were performed in an RNase-free environment in line with the manufacturer's recommendations and user guides. The brain was dissected immediately without PBS or PFA perfusion, and fresh frozen directly within an OCT mold in an isopentane bath submerged in liquid nitrogen. The lateral parts of the brain were trimmed during cryosectioning to allow 10-μm-thick coronal sections including both dorsal hippocampi to be cryosectioned into the capture area (6.5 mm by 6.5 mm area) of the Visium Gateway Gene Expression Slide (10x Genomics). Cryosections were taken at 1.9 mm posterior to bregma, >2.5 mm anterior to the injection site at 4.6 mm posterior to bregma. Cryosections were also taken adjacent to the sections used for Visium and from MEC to confirm increased ipsilateral GFP and c-Fos immunostaining respectively (data not shown).

The Visium Spatial Tissue Optimization Reagents Kit (10X Genomics) was used to determine an optimal tissue permeabilization time of 12 min when performed on 10-μm sections. Samples were mounted directly onto the Visium Gateway Gene Expression Slide and incubated at 37°C on a thermocycler (T100 Thermal Cycler, Bio-Rad) for 1 min, followed by methanol (>99.9%, 34860, Sigma-Aldrich) fixation at −20°C for 30 min. Sections were blocked and stained with DAPI before coverslipping and imaging at 20× magnification using a Zeiss Axioscan 7 microscope. Next, tissue permeabilization was performed as previously to release polyadenylated mRNA for capture by poly(dT) primers on the slide. Steps for reverse transcription, second strand synthesis, and denaturation were followed according to the manufacturer's kit to produce cDNA fragments, from which spatially barcoded Illumina-compatible libraries were constructed using the dual Index Kit TT Set A (10X Genomics) to add unique i7 and i5 indexes for barcoding.

Single Cell 3′ v3.1 Dual Index libraries were sequenced on a NovaSeq SP instrument by Edinburgh Genomics using parameters recommended by 10X Genomics. These libraries were then demultiplexed by the sequencing provider using the mkfastq command distributed within 10x Genomics' Cell Ranger software (*105*). The resultant FASTQ was processed in-house by 10x Genomics' Space Ranger (v1.3.1). The fluorescent microscopy images had to be first manually aligned using 10x Genomics' Loupe Browser (v6.0.0). Next, ImageJ (v1.53f51) was used to convert images to a TIFF file at the appropriate dimensions. Then within Loupe Browser, an overlay was manually aligned with the fiducial frames present on the Visium slides. Finally, spots for downstream analysis were manually selected, excluding those where tissue was not fully present or was damaged. This produced an alignment file necessary for Space Ranger, which was then run with default parameters. Space Ranger produced a spot-by-count matrix suitable for downstream analysis along with summary statistics confirming successful sequencing and preprocessing. Downstream analysis was carried out in R (v4.0.2) with code available at https://github.com/eturkes/crowley-visium-EC-stim. The Seurat package (v3.2.2) (*106*) was used for all object handling. After reading the count matrix into Seurat, genes completely unexpressed in any spot were excluded. The counts were then normalized using SCTransform (*107*), as recommended by the Seurat authors for handling biologically expected variation in counts due to tissue anatomy. The aim of SCTransform is to maximize biological variation and minimize variation due to technical artifacts by combining information across genes and building a negative binomial distribution model. Both samples were normalized separately for exploratory analysis but normalized together when compared in quantitative analysis.

To identify DEGs in areas of interest, Loupe Browser was used to manually select spots for comparison between the ipsilateral and contralateral hemispheres. Comparisons included spots in the DGML and the hippocampus overall. Barcodes corresponding to the spots were exported as a CSV file for use in R. After subsetting the data to these spots, SCTransform was applied within the subset, the subset spots were pseudobulked by mouse of origin, and limma (v3.46.0) (*108*–*110*) was used to identify genes differentially expressed between the ipsilateral and contralateral hemispheres. Limma applies t-statistics in an optimized manner for sequencing data by borrowing information across genes to moderate technical artifacts such as mean-variance trend. The pooling of information across genes also allows null hypothesis with limited sample size, allowing us to use each mouse as true biological replicates rather than a pseudoreplication confounded per-spot analysis. DEGs were defined as those exhibiting changes between comparison groups with a false discovery rate (FDR)–corrected $P < 0.05$. These genes and others were visualized using functions available within Seurat and Loupe Browser. Visium raw sequencing data can be found at https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE327095, and analysis code can be found at https://github.com/eturkes/crowley-visium-EC-stim.

Xenium spatial transcriptomics

PFA-fixed and sucrose-cryoprotected frozen mouse brains were cut to 10 μm coronal sections (approximately AP −1.94 mm) with a cryostat, after which they were placed on Xenium slides (PN3000941, 10x Genomics) at RT using PBS. Slides were stored at −70°C until probe hybridization. Probe hybridization and Xenium run were performed using Xenium Prime reagents (PN1000720 and PN1000661, 10x Genomics), Xenium Prime 5K Mouse Pan Tissue & Pathways Panel (PN1000725, 10x Genomics) and mouse 51–100 gene add-on panel (PN1000766, panel ID Q43N8N, 10x Genomics) according to the manufacturer's instructions and following protocols CG000760 and CG000584, respectively.

Xenium spatial transcriptomics data from mouse brain tissue sections were processed using ScanPy (v1.9+), Squidpy, and SpatialData. Quality control metrics included total transcript counts per cell, number of unique genes detected, cell area, and nucleus-to-cell area ratio; outlier detection was performed using z-score transformation ($|z| > 3$) for each metric, with additional thresholds requiring a minimum of 300 genes per cell and 300 cells per gene. Doublet detection and removal were performed using Scrublet (*111*) with StellarPath (*112*) similarity. Expression data were normalized using library-size normalization (target sum = 100) followed by log1p transformation; highly variable genes ($n = 1000$) were identified in a batch-aware manner, and principal components analysis (PCA) was performed using 50 components with the ARPACK solver. A k-nearest neighbor graph was constructed using cosine distance with 200 neighbors and all 50 principal components. t-distributed stochastic neighbor embedding (t-SNE) was computed using cosine distance for visualization. Unsupervised clustering was performed using the Leiden algorithm (igraph implementation) with resolution parameter 0.20, selected from multiple resolutions (0.15 to 0.30) based on cluster granularity. Cell type annotation combined Gene Set Variation Analysis (GSVA) via Decoupler on pseudobulk expression profiles with differential marker analysis per cluster; reference marker gene sets were obtained from Azimuth Cell Types 2021 (*113*) and Allen Brain Atlas 10x scRNA 2021 (*114*), supplemented with curated B cell markers (*115*, *116*). Plasma cells were identified using a two-tier classification based on *Igkc* expression: high-confidence "PC_Gold" reference cells (*Igkc* count > 5) were used to validate "PC_Candidate" cells (*Igkc* count ≥ 1) via expression profile similarity, with PC_Gold cells and similar candidates classified as true positive plasma cells (PCTP) and the remainder as false positives (PCFP). Spatial autocorrelation of gene expression was assessed using Moran's I statistic (100 permutations), and spatial subclustering within each cell type was performed using MiniBatchKMeans (k = 2 to 6) with silhouette score optimization on spatial coordinates, accepting subclusters only when scores exceeded 0.65. Two complementary approaches identified distance-dependent genes to determine transcriptomic changes relative to plasma cell proximity: Pearson correlation identified genes with linear expression gradients relative to plasma cell distance, while NicheDE (*117*) identified genes with expression changes in close proximity. Cell type–specific distance cutoffs were computed as the mean of per-gene changepoints across significantly distance-dependent genes, dividing each cell type into "close" and "far" subpopulations relative to PCTP. Differential expression analysis (two-sided Wilcoxon rank-sum test, Benjamini-Hochberg FDR correction, adjusted $P < 0.05$) and pathway enrichment analysis [Enrichr (*118*) via gseapy; GO Molecular Function 2023, GO Cellular Component 2023, SynGO 2024 (*119*), Reactome Pathways 2024 (*120*), WikiPathways 2024 Human (*121*)] were performed across cell types, spatial subclusters, proximity-based subpopulations, and ipsilateral versus contralateral hippocampus. Raw Xenium sequencing data and analysis code can be found in Zenodo (*122*) and GitHub (https://github.com/ElectrOmics/Ger_Plasma_Xenium_scRNA_2025).

BCR-seq library preparation and analysis

Animals were deeply anesthetized and perfused with 20 ml ice-cold PBS as previous. Dural meningeal, hippocampal, and striatal tissue were collected and immediately placed in RNALater (Invitrogen, AM7020) to preserve RNA integrity. Samples were stored at −80°C until processing. Tissue was homogenized and total RNA extracted using Monarch Total RNA Miniprep Kit (New England Biolabs, T2010S) following manufacturer's protocol. Reverse transcription and BCR enrichment were performed using NEBNext Immune Sequencing Kit (Mouse) (New England Biolabs, E6330S) in accordance with the manufacturer's user manual. BCR-seq libraries were sequenced by Azenta Lifesciences using a 300 bp paired-end sequencing run. The generated BCR-seq data was processed using MiXCR (v4.7.0) (*123*) using the preset neb-mouse-rna-xcr-umi-nebnext.

Data analysis was performed in the R programming environment (v4.4.1). Clonotype data (considering only heavy chains) from MiXCR were analyzed using the R package immunarch (v0.9.1) (https://CRAN.R-project.org/package=immunarch). Number of unique clonotypes (defined using CDR3 amino acid sequences) for each tissue were extracted, overlaps calculated using the overLapper function in the package systemPipeR (v2.10.0) (*124*) and upset plots generated with UpSetR (v1.4.0) (*125*).

For BCR network analysis, the sequencing data were first processed using the nf-core/airrflow (*126*) workflow. The resulting clones file were then filtered to remove non-heavy chains and then passed to dandelion (v0.3.8) (*127*) to generate BCR network plots. BCR raw sequencing data can be found at https://www.ebi.ac.uk/ena/browser/view/PRJEB111235 and analysis code at https://github.com/clatworthylab/BCRseq_Alzheimer_UCL_Soyon.

Confocal imaging

Brain sections for direct comparison were imaged consecutively during the same imaging session. The same settings including laser intensity and pixel size were kept constant for all sections in the same immunostaining. Typically, a frame size of 1024 pixels by 1024 pixels was used for most experiments. Mounted slides of mouse brain sections were imaged in a single Z-plane using a Plan-Apochromat 10x/0.45 numerical aperture (NA) objective lens initially using the tile function on a Zeiss laser scanning microscope 800 (LSM800) to gain an overview of the mCitrine/EGFP expression, HA immunostaining and DiD Neuropixels 1.0 probe shank labeling at the MEC and hippocampal areas. For c-Fos and pPDH imaging, a Photometrics Prime camera attached to a Zeiss LSM800 microscope was used to acquire widefield images of the MEC region at 10x magnification in two or three brain sections per hemisphere. For RNAscope ISH experiments, entire DGML imaging of microglial P2Y12 and *C1qa* mRNA, or entire hippocampal imaging of *Igha*/*Igkc*/*Tnfsf13*/*Tnfrsf17* signal, was performed with a 20x/0.8NA objective, with 3 μm deep Z-stacks (1 μm step size) acquired per brain section and one or two sections analyzed per hemisphere or per brain for each animal. For measuring IgA and IgG staining at the DG, Z-stack images of 14 μm depth (1 μm step size) per brain section were acquired at 10x magnification and two sections analyzed per hemisphere. For PECAM1, AQP4, CRABP2 and IgG staining of the hippocampus, overview Z-stack images of 30 μm depth per brain section were acquired at 20x magnification with high-resolution images acquired at 93x using a Leica Stellaris 8 microscope. For P2Y12 and CD68 imaging at the DGML, 3 Z-stack images of 7 μm depth (0.5 μm step size) per brain section were acquired at 20x magnification using a Zeiss LSM800 microscope and 2 sections analyzed per hemisphere. For Iba1 and CD11b imaging, 8 μm Z-stack images (0.5 μm step size) were acquired from the DGML using a 40x/1.3NA objective of a Zeiss LSM800 microscope. High-resolution microglial P2Y12 and CD68 images were acquired using the 63x/1.4NA objective of a Zeiss LSM800 microscope. Frame size of 2048 by 2048 was used for these images. Eleven-micrometer Z-stacks were imaged with a voxel size of 0.05 μm by 0.05 μm by 0.220 μm. A minimum of three microglia were imaged per brain section and three sections analyzed per hemisphere. Acquired data was analyzed either on ImageJ Fiji or Imaris software. Representative images are obtained either from ImageJ Fiji or Imaris and are adjusted for brightness, contrast, and background where appropriate.

ImageJ Fiji analysis

To quantify c-Fos and pPDH immunoreactivity, widefield images were background subtracted using a 50-pixel rolling ball radius followed by binarization by threshold. Total c-Fos immunoreactive area per ROI was measured and values averaged per hemisphere. To quantify P2Y12, CD68, Iba1, and CD11b immunoreactivity, maximal intensity Z-projections were generated and binarized by threshold, with the % area covered then measured. To quantify IgA and IgG immunoreactive area, maximum intensity projections of DG images were generated in ImageJ followed by thresholding and binarization of immunopositive signal. The % area covered by IgA or IgG was then measured and averaged across sections from the same hemisphere.

Imaris analysis

For 3D reconstruction of microglia, Imaris (v10) 3D "Surfaces" function with "Shortest Distance Calculation", "Absolute Intensity" and grain size of 0.1 μm were used. 3D surface rendering was used to generate a surface on the P2Y12 channel. Next, the CD68 channel was masked to the P2Y12 surface to obtain only CD68 signal within the microglial cell being analyzed. A surface was then created on the CD68 masked channel with a volume filter of >0.01 μm^3 applied. Total P2Y12 volume per cell and CD68 volume per cell were measured and quantified per mouse hemisphere.

Airyscan confocal imaging and analysis

For imaging of synaptic markers, C1q, 4G8+6E10, VGLUT expression at EGFP-positive axon terminals in DGML, and IgG/B220 cells at GLUT1 vessels, the Airyscan function of a Zeiss LSM880 microscope with Plan-Apochromat 63x/1.4 NA oil-immersion objective lens with a zoom factor of 1.8 was used. Optimal frame size and Z-step distance for 3D stacks was determined by applying Nyquist sampling theorem (approximated to 2.3x) to the information frequency to be measured, i.e., axial resolution = ~400 nm for 488 nm laser ≥ 400/2.3 = 173 nm Z-step size. Eight Z-steps were acquired per stack, meaning a stack thickness of 1.21 μm when 488 nm was the shortest wavelength acquired and 1.47 μm when 594 nm was the shortest wavelength acquired (0.21 μm Z-step size based on Nyquist sampling of 594 nm wavelength). Three adjacent image stacks (regions of interest; ROIs) were acquired for a given brain region in a given brain section to ensure uniformity of the signal. Deconvolution of Airyscan images was performed using Airyscan processing at strength 6.0 on Zen Black software (v3.0). Deconvoluted output files from Airyscan images were converted to .ims files using Imaris File Converter (v9.5.1). Image files were given pseudonyms to blind from genotype and/or treatment group.

For analysis of synaptic markers and C1q density, .ims image files were loaded into Imaris software (v9.5.1, 10). Channel brightness was adjusted to optimize visualization of immunopositive puncta. The "Spots" function was used with "Region Growing," "Shortest Distance Calculation," and "Background Subtraction" functions enabled to select puncta of size 0.2 μm XY diameter and 0.5 μm Z diameter, values based on actual PSF calculations made using sparse fluorescent bead imaging on the Zeiss LSM880 microscope used for these experiments. A threshold was applied to immunoreactive puncta using the "Intensity Center" filter. Threshold values varied between channels and immunostainings but were kept consistent between image pairs being directly compared and between images acquired during the same imaging session. Spot size boundary was defined by "Local Contrast" using a value that appropriately reflected the immunoreactive signal of the source channel. Spots were then filtered for size, where spot volume ≥ 0.007 μm^3 based on the minimum theoretically resolved spot size using Airyscan of 0.14 μm by 0.14 μm by 0.35 μm = 0.007 μm^3. For colocalization of Bassoon and Homer1 as a measure of putative excitatory synapse density, a matrix laboratory (MATLAB) script ("Colocalize Spots" XTension) was implemented to count presynaptic and postsynaptic spots colocalizing with ≤250 nm separation between spot centers (*128*). The average value of colocalized spots obtained from a single image was divided by the total image volume to provide an estimate of total synapse density for the ROI. To calculate VGLUT1 and VGLUT2 coverage on GFP-positive axons in the DGML, axons were surface rendered based on the GFP channel. Next, VGLUT1 and VGLUT2 puncta were modeled using the spots function as above. The "Spots Close to Surface" XTension was then used with the shortest distance separation set to 0.0 to determine the number of VGLUT1 and VGLUT2 puncta overlapping with the GFP-positive axon surface.

Statistical analysis

Mice included in all analyses were age- and sex-matched littermates wherever possible. WT DREADD surgeries were performed in randomized sets of 4 littermates with EGFP+Veh, EGFP+CNO, hM3Dq+Veh and hM3Dq+CNO treatments performed simultaneously. Sample size was determined by applying power calculations to preliminary results from at least $n = 3$ biological replicates. All analyses were performed blinded to genotype and/or treatment group. Statistical testing was performed on R (v.4.3.2) and GraphPad Prism 10.0 (GraphPad Software). Statistical tests have been performed per experimental/biological replicate, either per animal or per brain hemisphere. Data points shown are animal averages or hemisphere averages (animal/hemisphere average = average of all brain sections, may vary slightly from ROI averages as a result) where points are linked in DREADD experiments. Data shown as mean ± standard error of the mean (SEM) throughout. Statistical tests and data representation are stated in the figure legends. Raw data were tested for outliers using ROUT (Q = 1%) analysis. Residuals were tested for normality using a Shapiro-Wilk test and tested for homogeneity of variance by either F test or Levene's test depending on the data type. Data that failed either test were $\log_{10}$ transformed followed by retesting. Where a two-way mixed analysis of variance was used, "treatment group" or "brain region" were used as the between-subjects factor and "hemisphere" was used as the within-subject factor. Significant interaction effects were followed by pairwise t tests comparing contra- and ipsilateral hemispheres, Bonferroni-adjusting the P values for multiple testing. Differences between two groups were compared using paired or unpaired t tests as indicated, or alternatives such as Welch's t test or Wilcoxon matched-pairs signed rank test where assumptions for parametric testing were not met. Throughout, $^{ns}P > 0.05$, $*P < 0.05$, $**P < 0.01$, $***P < 0.001$, $****P < 0.0001$. Detailed statistical data can be found in tables S2 and S3. ROI data per graph can be found in table S4.

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SUPPLEMENTARY MATERIALS

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C1q and immunoglobulins mediate activity-dependent synapse loss in the adult brain

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Editor's summary
Loss of synapses is a major hallmark of Alzheimer's disease (AD) and other neurodegenerative disorders. In AD-relevant disease models, deposition of C1q, the central protein in the classical complement cascade, contributes to synapse loss. Crowley *et al.* showed that neuronal hyperactivity triggers C1q protein deposition at synaptic targeting sites accompanied by accumulation of B cells, resulting in the loss of synaptic components (see the Perspective by Miarka and Prinz). In an AD model already displaying neuronal hyperactivity, dampening the hyperactivity lowered C1q deposition and partially restored loss of synaptic components. Intriguingly, B cell–derived immunoglobulin M participated in activity-dependent synapse loss. These findings provide evidence of neuronal activity as a regulator of C1q-mediated synapse loss in the adult brain and introduce immunoglobulins as key players in this process. —Mattia Maroso

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